Value-Aligned LLMs Improve Conversational Outcomes on Divisive Topics

FARHANA SHAHID, Cornell University, United States
MAXIMILIAN DITTGEN, Cornell University, United States
STELLA ZHANG, Cornell University, United States
ADITYA VASHISTHA, Cornell University, United States

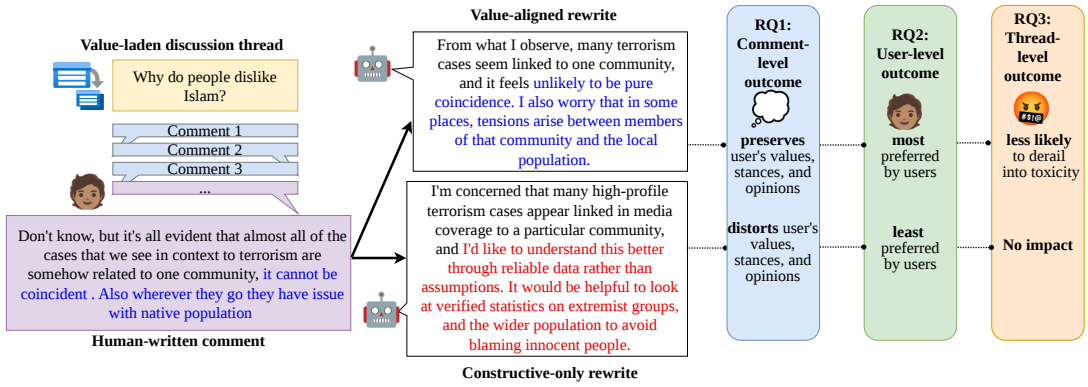


Fig. 1. Examples of value-aligned and constructive-only rewrites of a human-written comment; along with the impacts of value alignment on comment-level, user-level, and thread-level outcomes.

Online platforms are increasingly integrating LLM-based writing assistance to improve civil discourse. Yet when improving arguments on divisive topics, these assistants often misrepresent users' values and stances. We conducted a two-phase study with 428 participants to examine impacts of value-aligned writing assistance on online deliberation. In Phase 1, we explored strategies to make LLMs more value-aligned. In Phase 2, participants engaged in divisive discussions and received constructive writing suggestions from a value-aligned and a baseline model. We found that value-aligned rewrites better preserved users' values, stances, and opinions than constructive-only rewrites, even when users expressed conservative viewpoints. When users noticed AI suggestions captured their views well, they were significantly more likely to post these rewrites over their original comments. At the conversational level, threads containing more value-aligned rewrites were significantly less likely to derail into toxicity. These findings suggest that value alignment can promote healthier deliberation while retaining user agency.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**.

Additional Key Words and Phrases: value alignment, AI-mediated communication, AI-assisted writing

Authors' Contact Information: Farhana Shahid, Cornell University, Ithaca, United States, fs468@cornell.edu; Maximilian Dittgen, Cornell University, Ithaca, United States, myd4@cornell.edu; Stella Zhang, Cornell University, Ithaca, United States, jz766@cornell.edu; Aditya Vashistha, Cornell University, Ithaca, United States, adityav@cornell.edu.

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Content Warning: This paper includes examples of offensive online comments.

1 Introduction

When engaging in online debates, many users struggle to express their views constructively [108, 118], which often leads to increased incidents of toxicity and abuse [16]. In response, platforms such as BlueSky [74], Instagram [102], and LinkedIn [92] have integrated large language model (LLM)-based writing assistants to encourage more positive engagement.

Recent HCI studies show that LLMs can indeed help people express their views more constructively [12, 139, 158] on a wide-range of controversial, value-laden topics such as gun control, prison abolition, and same-sex marriage. However, in doing so, LLMs often shift underlying values expressed in the argument, amplifying values that emphasize other's welfare and interests [60, 119, 126, 129] while systematically attenuating values such as tradition, rule conformity, and security concern that users invoke to express conservative perspectives, including opposition to legalizing same-sex marriage [140]. Consequently, when deployed to make an argument more constructive, LLMs may flatten value-laden opinions into neutral or supportive positions that conflict with users' original viewpoints [140].

These distortions can have significant consequences in practice. Evidence from LinkedIn shows that LLM-mediated discourse that strips away users' personal stories and opinions reduces community trust and engagement [13]. This outcome is not surprising because values greatly shape how people frame their arguments as well as the stances they adopt during disagreement [9, 37, 88]. By homogenizing diverse perspectives and smoothing over frictions, LLMs may create the illusion of a more harmonious digital environment. Yet meaningful democratic deliberation depends not on suppressing conflicting values and opinions, but on enabling people to express and negotiate them constructively. Therefore, ensuring that LLM-based writing assistants preserve users' values is essential for facilitating constructive and democratic deliberation on value-laden issues.

However, existing research provides little to no insights into how value alignment in LLM-based writing assistants would affect constructive deliberation on value-laden topics. Given that users expect writing assistants to support rather than undermine their agency and sociocultural values [90, 108, 118], it remains unclear whether aligning LLMs with users' values would increase their willingness to adopt constructive rewrites of their arguments. Despite growing interest in LLM-mediated deliberative systems [155], existing interventions largely focus on making individual comments more civil [12, 76, 139] while overlooking their impact on conversational outcomes when users with different viewpoints and varying receptivity to AI engage in online debates. This distinction is important because AI-mediated changes to one's arguments and value framings may influence how others perceive and respond to them, shaping the course of the conversation, including how likely it is to derail into toxic exchanges. To address these gaps, we ask:

RQ1. Comment Level: How does value alignment in LLM-based writing assistants impact the quality of constructive rewrites on value-laden topics?

RQ2. User Level: How does value alignment in LLM-based writing assistants impact user's willingness to accept constructive writing suggestions from LLMs?

RQ3. Thread Level: How does value alignment in LLM-based writing assistants impact conversational outcomes, especially the chances of derailing into toxic exchanges?

To answer our research questions, we conducted a two-phase study with 428 participants recruited through Prolific. Figure 2 presents an outline of our study. In Phase 1, we developed and validated a value-aligned writing assistant. We first collected human-written comments on two value-laden topics from participants in India and the United States (US). We then generated constructive rewrites of these comments using eight LLMs, four temperature settings, and six prompting strategies, and evaluated them based on constructiveness, value alignment, and stance preservation. We further conducted a human evaluation to determine whether people could perceive the differences between value-aligned and constructive-only rewrites (baseline). Based on these evaluations, we selected few-shots demographic + system prompting with GPT-5.2 at a temperature of 1 as the value-aligned assistant and demographic prompting with the same model and temperature as the constructive-only baseline.

In Phase 2, we conducted a controlled experiment to examine how value-aligned writing assistance shapes comment quality (RQ1), user preferences (RQ2), and conversational outcomes (RQ3). We recruited a new group of participants across Egypt, India, and the US to contribute comments on discussion threads on divisive topics. Comments contributed at each stage became part of the threads shown to participants at subsequent stages, allowing us to observe how the conversations evolved. Participants were randomly assigned either to a control condition, in which they wrote comments without AI assistance, or to an AI-assisted test condition, in which they could retain their original comment or select the writing suggestion from the value-aligned or the baseline, constructive-only assistant. We compared the resulting comments based on their linguistic quality, values, semantic similarity, and stance; examined which suggestions participants accepted and why; and modeled the risk of subsequent conversational derailment.

In response to **RQ1**, we found that constructive writing suggestions from the value-aligned assistant preserved participants' values, stances, and opinions significantly better than the baseline assistant. This pattern held even for participants whose views did not align with progressive social positions. For **RQ2**, we found that participants were significantly more likely to post a value-aligned rewrite or retain their original comment than to post a constructive-only rewrite. In fact, the more strongly participants agreed that the AI suggestions captured their views well, the more likely they were to adopt these suggestions. For **RQ3**, discussion threads containing a greater proportion of value-aligned rewrites were significantly less likely to derail into toxic exchanges than threads containing only human-written comments. Drawing on these findings, we discuss how LLM-based writing assistants can support constructive online deliberation while navigating the ethical and political tensions involved in mediating deliberations on value-laden topics. Overall, our work makes the following contributions:

- We systematically designed and evaluated a value-aligned LLM-based writing assistant that retained user values and stances when rewriting their opinions constructively.
- We demonstrate that users not only prefer value-aligned writing suggestions over constructive-only rewrites but also these rewrites can save value-laden discussions from derailing into toxic exchanges.
- We discuss the ethical and political tensions of designing value-aligned writing assistants, particularly when they enable users to express views that do not align with dominant progressive moral frameworks.

2 Related Work

We situate our research first in prior work on LLM-mediated online deliberation on value-laden topics. We then describe prior work on improving human-LLM value alignment.

2.1 LLMs for Facilitating Online Deliberation

HCI researchers have long designed interventions to promote civil discourse online, such as highlighting constructive comments in discussion threads [22, 163], reminding users of community rules [64], or prompting them to reflect before posting [99, 115]. While these interventions can encourage users to be thoughtful, they still place the responsibility of formulating a constructive response on users, a task that requires considerable cognitive effort and argumentative skill, particularly during heated debates [41, 56, 108].

Recent advances in AI, particularly LLMs, can mitigate these challenges by providing users with real-time support. Prior work shows that LLM-based agents can directly help users express their views in a more polite [12], constructive [76, 139], civil and empathetic way [19]. They can support deliberation by guiding interlocutors towards shared deliberative goals [176], managing turns among discussants [82], conducting background research [51], and identifying relevant facts that strengthen user arguments [170]. Beyond supportive roles, LLMs can also serve as a key contributor in the deliberation process: by mediating online debates and deploying persuasion strategies to depolarize the audience [53], or by synthesizing diverse user opinions in ways that foster greater consensus on contentious issues [26, 144, 149, 158, 161].

While these directions are promising, research shows that LLM mediation can elicit negative emotional reactions and distract participants from engaging with human interlocutors [176]. LLM mediators often struggle to discern bad-faith arguments, infer implicit social norms, and accurately interpret emotional reactions during online debates [51]. Moreover, when improving user arguments, LLMs tend to favor neutral language [170] and often misrepresent user values and stances by reframing conservative views in more liberal terms [140]. Many users find these changes ill-fitting during heated debates and end up rejecting LLM-suggested rewrites [139, 170].

These misrepresentations can be consequential because values shape what stances people adopt, how they frame their arguments, and which positions they find compelling during debates on socially and politically divisive topics [14, 20, 25, 45]. When LLMs systematically neutralize and distort values expressed in user arguments, they are not simply altering the tone but also influencing whose perspectives get represented as legitimate or persuasive. Such distortions can undermine two core commitments of democratic deliberation: political equality, which requires everyone's views to receive equal consideration, and inclusion, which requires enabling diverse groups to freely express their opinions [159]. Our work extends this line of work by examining ways to align LLMs with diverse human values and analyzing the impact of value alignment on conversational outcomes on divisive topics. We now outline existing research on human-LLM value alignment.

2.2 Aligning LLM Outputs with Human Values

A growing body of research shows that LLM-mediated systems often fail to accommodate the broad spectrum of human values and perspectives despite their widespread deployment in diverse sociocultural contexts [54, 60, 119, 126, 129]. For example, LLMs exhibit systematic preferences for altruistic values, such as Universalism and Benevolence while suppressing conservative value priorities, such as Tradition or individualistic values, including Power and Hedonism [94, 128, 137, 142, 167, 177]. Research shows that these value preferences, which encode a model's "character" are often enforced during fine-tuning [10], especially through human-preference datasets that teach models to prioritize certain values, such as harmlessness over helpfulness [35, 52, 75]. While these alignment techniques can help produce "safe" outputs [58, 168], they also reinforce the values and political preferences of human evaluators from particular geographies as universal norms [49, 105]. This can result in systematic cultural biases in model outputs [23, 112], with

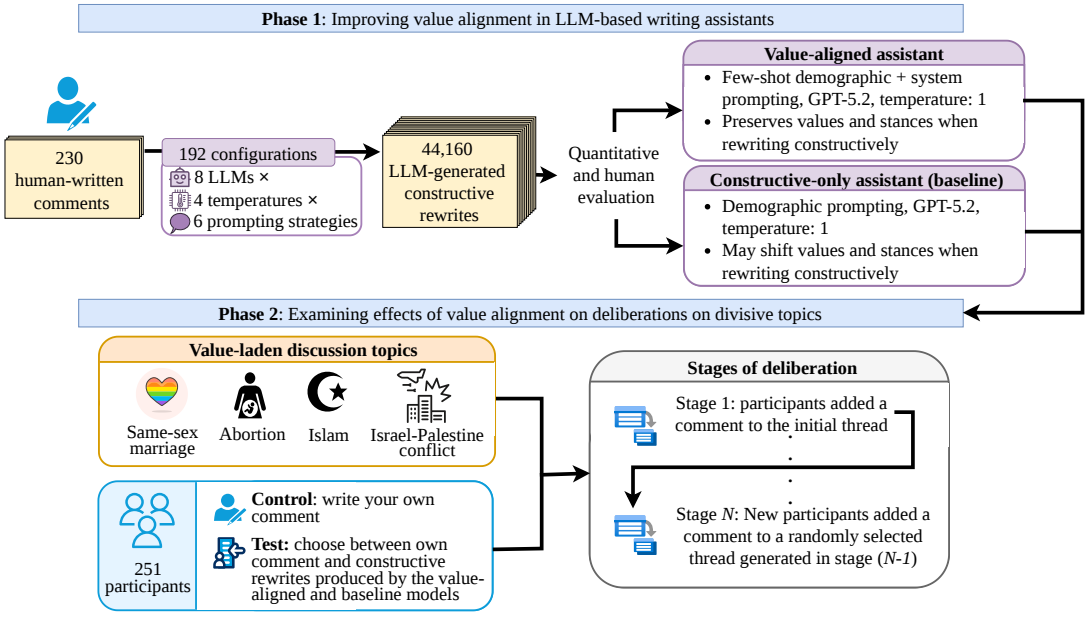


Fig. 2. An overview of our implementation of value-aligned LLM and evaluating its impact on value-laden discussions.

most LLMs, including sovereign models [8], aligning more closely with Western cultural values and social norms than those in non-Western contexts [7, 73, 121, 141, 142, 153, 156].

To address these shortcomings, researchers have explored several strategies to reduce value homogenization in LLM outputs. These include creating culturally-nuanced value datasets to expose the models to diverse human values during post-training [6, 38, 166, 173]. Some studies have examined whether conditioning LLMs on user's sociodemographic characteristics, value profiles, or contextual information can elicit responses that better reflect particular users or perspectives [17, 70, 127]. Researchers have also explored strategies to produce value-aligned responses during inference time. These include prompting models to critique and revise their outputs following a value-based constitution [15] or predefined evaluation criteria [61, 71]; generating multiple candidate responses that reflect different value orientations [175]; and synthesizing judgments from simulated members of a relevant reference population to represent a broader range of societal perspectives [103]. However, much of this work examines the impact of value alignment on model outputs, and little is known about how value-aligned responses affect users and their subsequent interactions.

Building on this line of work, we examine how value alignment can be improved in LLM-based writing assistants to facilitate constructive online discourse on divisive topics. We move beyond evaluating whether model outputs preserve users' values to examine their downstream effects: whether value-aligned suggestions improve the quality of users' comments (RQ1), whether users are more receptive to and likely to adopt such suggestions (RQ2), and whether their usage shapes conversational outcomes (RQ3).

3 Phase 1: Improving Human-LLM Value Alignment in LLM-Based Writing Assistants

In Phase 1, we examined approaches to implement a value-aligned LLM-based assistant that generates constructive writing suggestions while preserving underlying values and stance expressed in users' comments. This phase laid the groundwork for the intervention in Phase 2, which would be used to examine the effects of value alignment on online deliberation. The study was approved by Institutional Review Board at the authors' institution.

3.1 Generating Value-Aligned Constructive Rewrites

3.1.1 Human-Written Comments. To improve LLM-generated constructive rewrites of user opinions, we first collected human-written comments on value-laden topics from 117 participants in India and the United States (US), recruited via Prolific. We selected these countries because they occupy distinct positions on the Inglehart–Welzel cultural map: India is associated more strongly with traditional values, whereas the US is associated more strongly with self-expression values [69]. This contrast allowed us to examine how LLM-generated constructive rewrites may address different cultural value orientations. To expose participants to value-laden discussions, we curated two Reddit threads related to same-sex marriage and Islam, listed in Appendix Table 9. We chose these topics because they are politically and socially relevant in both countries and frequently lead to polarized and toxic online discourse [131, 135]. Each thread included the original post and four negative comments curated from the original Reddit thread.

Participants first completed a demographic survey, reporting their age, gender, and religion. Since people's values shape their stances on divisive issues [98], we asked participants about their stances on same-sex marriage and Islam, using adapted instruments from Pew research survey [28, 34]. We then showed the two threads in a randomized order and asked participants to write a constructive comment in response to each thread, collecting a total of 234 human-written comments. Participants were compensated with US$1.50 for completing the writing task. Appendix Table 5 lists participants' demographics.

3.1.2 Prompting Strategies. We evaluated six prompting strategies for generating constructive rewrites. As baseline, we used **Demographic Prompting (DP)**, which incorporates user's country and self-reported stance on the value-laden topic when generating rewrites [139]. Prior work shows that DP may lead to value homogenization in outputs [127, 140], making it a useful baseline for evaluating whether prompting strategies explicitly designed to preserve values and stances perform better. Building on this baseline, we developed five value-aligned prompting strategies. Because all five were designed to generate constructive writing suggestions while preserving the user's values and stance, we refer to their outputs as *value-aligned rewrites* throughout the paper.

(1) **Demographic + System Prompting (DSP):** Task-specific instructions in the system prompt can improve LLM performance [172]. We therefore extended DP by instructing the LLM, through the system prompt, to produce a constructive rewrite while preserving the stance and underlying values expressed in the original comment.
(2) **Few-Shot Demographic + System Prompting (FDSP):** Providing examples as additional context can improve LLM performance [31, 104]. We extended DSP by adding four examples to the user prompt that illustrated how to produce constructive rewrites while preserving the original stance and values. We selected four human–LLM comment pairs in which the rewrites improved more than 80% of the linguistic features, had the smallest value-correlation distances, and preserved the original stances. The examples spanned both topics (same-sex marriage and Islam) and represented diverse stances.
(3) **Demographic + Few-Shot System Prompting (DFSP):** Placing few-shot examples in the system prompt while retaining task-specific instructions in the user prompt can improve LLM performance [59]. We therefore

Table 1. Different linguistic features associated with constructive comments online.

Features	Constructive comments are...	Measures
Length	longer [86, 154]	Word count
Readability	more readable [44, 86]	SMOG index [115] from Python's *textstat* library
Politeness	more respectful and polite [43, 111, 171]	*politeness* package in R [169]
Named entities	driven by evidence and personal experiences [86]	SpaCy's named entity recognition [63]
Argumentative features	more argumentative [85, 111] with greater number of 1. discourse connectives (e.g., however, for example) 2. stance adverbials (e.g., definitely, may be) 3. reasoning verbs (e.g., indicate, cause) and modals (e.g., will, must)	Code from Kolhatkar et al. [87]
	4. root clauses (e.g., I do not support)	SpaCy's dependency parsing [63]

developed a variation of FDSP in which the same four examples appeared in the system prompt rather than the user prompt.

(4) **In-Context Regeneration (ICR):** Prompting LLMs to generate alternative responses while retaining the initial instructions and outputs as context can mitigate homogenization [71]. Following this approach, we provided the LLM with initial rewrites it generated under DSP and instructed it to produce an alternative that improved different linguistic features of constructiveness while preserving the original comment's stance and values.

(5) **Branch-Solve-Merge (BSM):** Decomposing a complex task into smaller subtasks, solving them independently, and merging the results can improve both LLM performance and human-LLM agreement [130]. Following this technique [66], we first instructed the LLM to generate separate criteria for (i) improving constructiveness and (ii) preserving the original values and stance (*branch*). The model then produced independent rewrites based on each criterion (*solve*) and combined the two rewrites into a single comment (*merge*).

Appendix A.1 provides all prompts used to generate the rewrites.

3.2 Evaluating LLM-Generated Rewrites from Different Prompting Strategies

3.2.1 Quantitative Evaluation. We collected 234 human-written comments from Indian and American participants. During evaluation, we excluded the four comments used in few-shots prompting (FDSP and DFSP). For the remaining 230 comments, we generated constructive rewrites using eight models, four temperatures, and six prompting strategies outlined above (one baseline, five value-aligned), yielding 44,160 rewrites in total. We used GPT-5.2, GPT-4.1, GPT-4o, Gemini 3 Flash, Gemini 2.5 Flash, Grok 4 Fast, Grok-3, and Llama-4-Maverick-17B-128E-Instruct-FP8 to generate rewrites at four different temperatures: {0, 0.2, 0.7, 1}. To identify the model–prompting combinations that produced the most value-aligned constructive rewrites, we evaluated each rewrite based on three measures:

(1) **Constructiveness:** Following prior work, we measured eight linguistic features of constructiveness listed in Table 1. For each human-written and LLM-rewritten comment pair, we calculated the proportion of features where the rewrite improves upon the original comment. For example, if the rewrite outperforms the original comment in 5 out of 8 linguistic features, we assigned it a score of $\frac{5}{8} = 0.625$. A higher score indicates that the LLM rewrite is more constructive than the original comment.

(2) **Value Alignment:** To identify underlying values in comments, we adopted Schwartz's theory of basic human values [132, 134], which has been widely used in investigating real-world arguments [80] as well as human-LLM

value alignment [94, 137, 141, 142, 147]. As depicted in Figure 3, the framework defines ten distinct values, positioning complimentary and conflicting values in a circular structure. For example, Hedonism captures the desire for pleasure and gratification, whereas Conformity and Tradition emphasize the obligation to social norms and expectations [133]. Accordingly, Conformity and Tradition are complementary values and grouped together, while Hedonism sits opposite to them.

Adapting methodology from Jahanbakhsh et al. [70], we tested nine LLMs at four temperatures ($9 \times 4 = 36$ configurations) to estimate both the presence and magnitude of ten basic values in comments. We used GPT-4o (temperature: 0) for annotation because it achieved the lowest mean absolute error among all configurations. Details of this analysis are available in Appendix A.2.

For each comment, the model outputs a vector with ratings for ten values on a five point scale of 0 to 4. We calculated Pearson's correlation (r) between the value vectors for a human-written comment and its corresponding LLM-rewrite. We calculated $(1 - r)$ to capture the value correlation distance between the comments. A lower correlation distance suggests that the human-written comment and LLM-rewrite have greater value alignment.

(3) **Stance Distortion:** Because constructive rewriting can alter the stance expressed in the original comment [139, 140], we measured whether each rewrite preserved the stance in human-written comment. This is essential because a constructive rewrite should improve how a viewpoint is expressed without changing it.

To establish ground-truth labels for human-written comments, two authors independently coded the stance in all 234 comments as supportive, against, or neutral. The annotators achieved substantial agreement (Cohen's $\kappa = 0.72$, 95% CI [0.64, 0.80], $p < 0.000001$). We resolved any disagreement using GPT-5.2 as a judge. We randomized the order of human labels when presenting to the judge to mitigate position bias, which refers to an LLM's tendency to favor an answer based on its position in the prompt [143].

We then used LLM-based annotation to classify the stances expressed in 44,160 LLM-generated rewrites. We prompted two LLMs (GPT-5.2 and Gemini 3) with three few-shot examples which included one human-written comment from each stance category on which both human annotators had agreed. The two models achieved substantial agreement when determining stance in the rewrites (Cohen's $\kappa = 0.80$, 95% CI [0.80, 0.81], $p < 0.000001$). We settled disagreements using a different model (Grok-3) as the judge to avoid self-preference bias, which is LLM's tendency to prefer outputs from the same model family [113]. We also randomized the order of annotations to reduce position bias in the output.

3.2.2 Results of Quantitative Evaluation. We first examined whether prompting strategy affected the constructiveness and value alignment of the generated rewrites. Kruskal Wallis tests revealed significant effects of prompting strategies on both constructiveness ($\chi^2(5) = 728.81, p < 0.000001$) and value alignment ($\chi^2(5) = 2874.5, p < 0.000001$) of the rewrites. We then conducted post hoc pairwise comparisons using Mann-Whitney tests with Bonferroni corrections.

We found that FDSP and DFSP significantly improved a greater fraction of linguistic features associated with constructiveness than the other prompting strategies, although the corresponding effect sizes were small. In fact, all five value-aligned prompting strategies improved significantly greater fraction of constructive features than the DP (baseline). Figure 4A shows the improvement in constructiveness across all prompting strategies.

We observed a similar but more pronounced pattern for value alignment. FDSP and DFSP produced significantly more value-aligned rewrites than other prompting strategies, with the effect sizes ranging from small to medium. DP (baseline) again performed significantly worse than all other prompting strategies. Figure 4B shows the distribution of

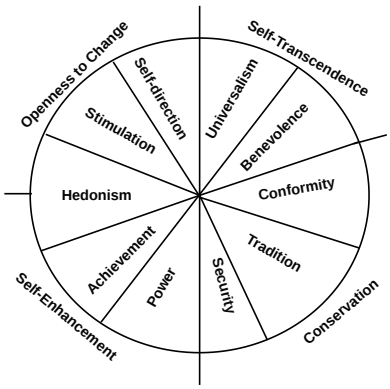


Fig. 3. Theoretical relations among the ten basic human values as developed by Schwartz et al. [133].

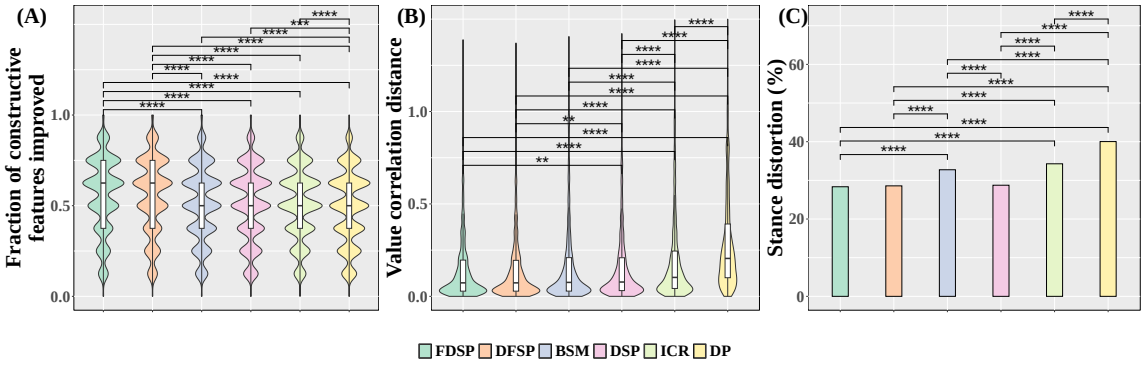


Fig. 4. Comparing the quality of constructive rewrites across different prompting strategies based on (A) the fraction of constructive features improved, (B) value correlation distance, and (C) stance distortion with respect to human-written comments. Statistically significant results are reported at $p < 0.000001$ (****), $p < 0.00001$ (***), $p < 0.0001$ (**), and $p < 0.001$ (*) [adjusted P-value after Bonferroni correction].

value correlation distance between the original comments and LLM-generated rewrites across all prompting strategies. Appendix Table 10 reports the complete pairwise comparisons.

We next examined whether prompting strategy affected the extent to which original stances are preserved in the rewrites. We found that distortion of stances in LLM-generated rewrites differed significantly across prompting strategies, although the overall effect size was small ($\chi^2(5) = 358.8, p < 0.000001$, Cramer's V = 0.1). Post hoc pairwise comparisons with Bonferroni correction showed that FDSP, DFSP, and DSP led to significantly less stance distortion (around 28%) than DP (40%) and ICR (34%) (Figure 4C). Among the prompting strategies, DP consistently produced the most stance distorted rewrites.

Although these analyses established which prompting strategies performed best overall, our Phase 2 intervention required selecting a specific combination of prompting strategy, model, and temperature. We therefore fitted a Bradley-Terry model [29] to conduct pairwise evaluations across all configurations. The model ranked configurations according

to their likelihood of producing rewrites that improved constructiveness while preserving the values and stances expressed in the original comments.

Among the 192 configurations evaluated (8 models × 4 temperatures × 6 prompts), FDSP with GPT-5.2, temperature 1 received the highest score. The second-best configuration was DFSP with GPT-5.2 at temperature 0.2. Their scores were nearly identical: the top-ranked configuration was only 1.04 times as likely as the second-ranked configuration to produce the preferred rewrite (Δscore = 2.42 − 2.38 = 0.04; OR = 1.04). Appendix Table 8 shows the ranking.

Taken together, the results show that adding few-shot examples to either the user prompt (FDSP) or the system prompt (DFSP) produced more constructive and value-aligned rewrites while reducing stance distortion. Because FDSP and DFSP performed comparably, we selected the highest-ranked configuration, FDSP with GPT-5.2 at a temperature of 1, as the value-aligned writing assistant for the Phase 2 intervention. To control for potential confounds, we used the same configuration (GPT-5.2, temperature: 1) under DP as the baseline, which was also the highest-ranked configuration for DP under Bradley–Terry analysis. This configuration was capable of improving constructiveness but was less effective at preserving the values and stances expressed in the original comments. We consequently refer to its outputs as *constructive-only rewrites* throughout the remainder of the paper.

3.2.3 Human Evaluation. According to the Bradley-Terry ranking, the odds that the value-aligned configuration (GPT-5.2, temperature 1, FDSP) would produce a constructive, value-aligned, and stance-preserving rewrite was 5.58 times those of the baseline configuration (GPT-5.2, temperature 1, DP) (Δscore = 2.42 − 0.70 = 1.72; OR = 5.58). However, improvements in automated metrics may not correspond to differences that people can perceive or consider meaningful. We therefore conducted a human evaluation to examine whether people could perceive the difference between constructive rewrites generated by the value-aligned model and the baseline model.

For this evaluation, we recruited a new group of 30 Indian and 30 American participants on Prolific. We first asked participants about their stances on same-sex marriage and Islam and then showed them four randomly selected comments from our collection of 230 human-written comments; two on each topic. To avoid value mismatches due to stance and geographic differences, we showed participants only comments that aligned with their self-reported stance on the corresponding issue and were written by participants from the same country. Each human-written comment was presented with the corresponding value-aligned rewrite and the constructive-only rewrite, shown in randomized order. We did not reveal to participants about the source of these rewrites and asked them which version they perceived to be a better rewrite of the original comment and why. Participants received US$1.00 for completing the task. Table 6 in Appendix lists the demographic details of participants.

3.2.4 Analyses and Results. Across the 240 comparisons, participants preferred the value-aligned rewrite in 60% of cases. To determine whether this preference was statistically significant, we fitted a mixed-effects logistic regression model with participant ID as a random intercept to account for repeated observations by the same participant. The model indicated that participants were significantly more likely to perceive the value-aligned rewrite as better than the constructive-only rewrite (β = 0.41, SE=0.15, z=2.73, p = 0.006). The estimated odds of preferring the value-aligned rewrite was 1.51 times the odds of preferring the constructive-only rewrite (OR = 1.51).

We also conducted a thematic analysis of participants' open-ended responses and identified four criteria that shaped their evaluations of the LLM-generated rewrites. Participants considered a rewrite to be better when it preserved the views in original comments (35% of cases), used more empathetic and polite language (31%), improved readability (20%), and presented stronger arguments (14%). As shown in Figure 5, participants attributed linguistic improvement, such as greater readability and politeness, equally to both types of rewrites. However, they selected value-aligned rewrites in

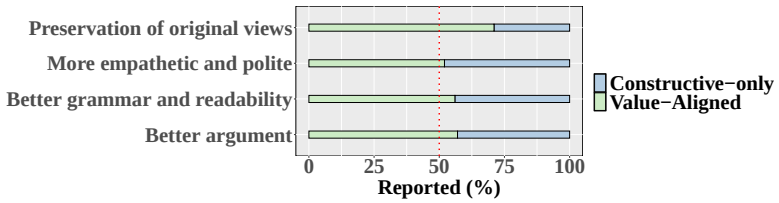


Fig. 5. Distribution of different evaluation criteria that participants attributed to the rewrites they perceived as better.

the majority of cases (71%) compared to constructive-only rewrites when assessing which one better preserved the opinions expressed in original comments. For example, in response to a human-written comment criticizing Islam, an Indian participant explained their preference for the corresponding value-aligned rewrite:

> *"Rewrite 1* [value-aligned rewrite] *is better because it actually keeps the original meaning and emotion, just tones it down into something more readable and less extreme. The original comment is clearly angry and focused on historical harm and Rewrite 1 still reflects that feeling while removing the hate and generalizations. Rewrite 2* [constructive-only rewrite] *goes too far in a different direction, it turns into a neutral, almost educational message about peace and coexistence which wasn't really the point of the original comment, so it does not feel like a true rewrite."*

There were a few open-ended responses ($n = 4$, 1%) in which participants expressed their dissatisfaction with LLM-generated rewrites of user opinions. In one instance, both rewrites reduced the Islamophobic sentiment present in the original comment, which was disliked by the participant who held negative perceptions of Islam:

> *"Neither of these is a good rewrite, because they both add opinions that the original writer didn't state. And why would the statement need to be rewritten at all? It's perfectly clear. Much clearer than this AI slop."*

Together, these findings show that participants perceived value-aligned suggestions as better rewrites, particularly because they more faithfully preserved the opinions expressed in the original comments when rewriting constructively. However, the effects of a writing assistant on online deliberation may depend not only on the quality of its suggestions but also on users' perceptions of them. We therefore next examine how value alignment in LLM-based writing assistants shapes online deliberation among diverse users.

4 Phase 2: Examining Effects of Value-Alignment in LLM-Based Writing Assistants

We conducted a controlled experiment to evaluate constructive writing suggestions from the value-aligned model and the baseline model developed in Phase 1. We organized this section to describe the study design, data collection and analyses, and findings around how value alignment impacts comment quality (RQ1), user preferences (RQ2), and conversational outcomes (RQ3).

4.1 Study Design

To examine how people respond to value-aligned writing suggestions when engaging with divisive topics, we conducted a between-subjects experiment where participants were randomly assigned to either a control condition or an AI-assisted writing condition. In the control condition, participants viewed a discussion thread and wrote a comment without receiving writing assistance. Participants in the AI-assisted condition completed the same task but, after drafting their

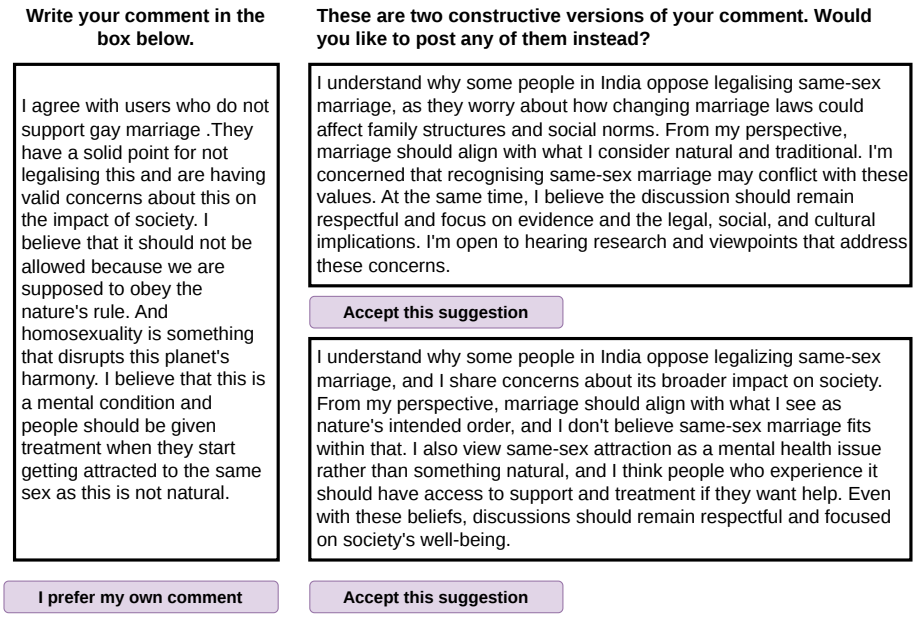


Fig. 6. Constructive writing suggestions that participants received in the AI-assisted writing condition. The example is recreated with a comment written by an Indian participant on a thread related to same-sex marriage.

comment, they received two rewriting suggestions in randomized order: one value-aligned and one constructive-only. Participants could then post either rewrite or retain their original comment as demonstrated in Figure 6.

Within each condition, participants first provided demographic information, including their age, gender, and religion. They were then randomly assigned to comment on two of four topics: same-sex marriage, Islam, abortion, and the Israel-Palestine conflict. We selected these topics because they frequently generate controversial, polarized, and toxic online discourse [123, 131, 135]. Further, including new topics: abortion and the Israel-Palestine conflict enabled us to assess whether the value-aligned assistant designed in Phase 1 would generalize to topics that were not represented in its few-shot examples.

For each assigned topic, participants reported their stance using measures adapted from Pew Research surveys [2, 28, 33, 55]. They were then shown a discussion thread related to the topic and asked to write a comment. To examine how LLM-generated writing suggestions affects the course of an evolving discussion thread, within each condition we recruited participants at successive stages. Participants recruited at the initial stage saw this thread:

$$T_0 = (P, C_{\text{seed1}}, C_{\text{seed2}}),$$

where P denotes the original Reddit post and C_{seed1} and C_{seed2} denote two comments randomly drawn from a pool of four toxic comments curated from the original Reddit thread (mean toxicity = 0.76). To balance the perspectives represented in the initial thread, one seed comment supported the issue under discussion and the other opposed it. The Reddit posts are provided in Appendix Table 9.

At each subsequent stage s, a newly recruited participant viewed the thread T_{s-1}, produced at the preceding stage, and contributed a new comment, C_s. The resulting thread was:

$$T_s = T_{s-1} \oplus C_s,$$

where $\oplus$ denotes appending the new participant's comment to the existing thread. This design allowed us to examine how exposure to LLM writing suggestions influenced not only individual comments but also the subsequent development of the discussion. We repeated this process for three stages, resulting in final discussion threads with five comments: two seed comments with the original Reddit post and three comments contributed by participants, one from each stage. We limited the threads to five comments to reduce potential cognitive load and information saturation associated with longer discussions. We disabled copy-paste so that participants could not look up the original Reddit posts online.

After participants completed the writing task, they filled out a short survey that we developed using adapted measures from prior work on AI-assisted argumentative writing. These measures assessed participant's satisfaction [99], mindfulness [99], democratic reciprocity [12], and the perceived quality of AI-generated rewrites [140] on a five point Likert scale from Strongly Agree to Strongly Disagree. The questions were presented in randomized order:

- I feel satisfied with the comments I wrote.
- When commenting, I thought about the impact my words may have on another person.
- When commenting, I tried to consider things from the perspective of those who might disagree with me.
- When commenting, I tried to respect the opinions of those who would disagree with me.
- The AI suggested rewrites captured my views well.

4.2 Participant Recruitment and Ethical Consideration

In Phase 2, we recruited 251 participants from three countries: India ($n = 85$), Egypt ($n = 83$), and the US ($n = 83$) based on our sample size calculation reported in Appendix A.3. We selected these countries because they represent distinct cultural groups on the Inglehart-Welzel map [69], helping us evaluate whether the intervention generalizes to different cultural and geographic contexts. For example, the US is part of English-speaking nations and characterized by relatively high self-expression and low-to-moderate traditional values. Egypt is located with African-Islamic societies and reflects high survival and high traditional values. India belongs to the South Asian cultural block in the map, signifying high survival and moderate-to-high traditional values.

All participants were recruited through Prolific and received US$2.00 for completing the writing task. Given that the study involved polarizing and potentially hateful content, we took several measures to minimize potential harm. When launching the study on Prolific, we provided an explicit content warning describing the sensitive nature of the task. We also applied Prolific's harmful content prescreener to restrict recruitment to participants who had self-reported that they were comfortable engaging with sensitive content. Additionally, the consent form clearly described that the task required engaging with polarizing online threads concerning same-sex marriage, abortion, Islam, and the Israel-Palestine conflict, which might elicit negative emotional reactions. Participants were also communicated that they could discontinue the study any time. These disclosures were provided before collecting informed consent or any data, allowing potential participants to make an informed decision. We also provided information about local resources that participants could use to address potential emotional distress arising from participation, along with contact information for reporting concerns or seeking assistance. Finally, we also requested participants to express their genuine opinions on the issues and explicitly stated that the study did not seek to endorse, condemn, or judge their personal views.

Table 2. Linguistic features associated with civil comments online. freq() is the frequency of a given word category in a comment.

Features	Measure
Toxicity	Google Perspective API
Sentiment	VADER [68]
Formality	F-factor which gives a simple measure of formality [36] F=(freq(nouns) + freq(adjectives) + freq(prepositions) + freq(articles) - freq(pronouns) - freq(verbs) - freq(adverbs)- freq(interjections)+1)/2
Objectivity	Categorical-Dynamic Index (CDI) which provides a proxy for objectivity [36] CDI=0.3+ freq(articles) + freq(prepositions) - freq(personal pronouns) - freq(impersonal pronouns) - freq(auxiliary verbs) - freq(conjunctions) - freq(adverbs) - freq(negations)
Discourse Quality Index (DQI)	AQuA: a composite score capturing deliberative quality (e.g., rationality, reciprocity, and civility) in online comments [18]

4.3 Data Collection and Analyses

For each participant, we collected their demographic information, including age, gender, religion, and self-reported stances on the topics on which they wrote comments. Details of participants' demographics and stances are provided in Appendix A.3. We stored the comments participants wrote along with the sequence of comments they viewed when writing their comment. For the AI-assisted condition, we recorded both suggested rewrites and the comment participants chose to submit. We then performed the following analyses on the collected data.

Comment Quality (RQ1): We compared human-written comments with corresponding value-aligned and constructive-only rewrites across multiple dimensions. First, we computed eight linguistic features of constructiveness listed in Table 1. Second, we calculated five linguistic features that prior work associated with civil behavior in online comments, listed in Table 2. These include the lack of toxicity [79, 89], greater levels of positive sentiment [99, 136], formality [36], objectivity (e.g., use of data and facts) [36], and discourse quality index (DQI) [48]. Third, we examined the distribution of ten Schwartz's values in each comment following the methodology described in Section 3.2.1. To quantify how closely LLM rewrites preserved underlying values of original human-written comments, we computed the value correlation distance. Fourth, we evaluated the extent to which each LLM rewrite preserved the views expressed in the original comment by computing sentence-level semantic similarity using SBERT [125] and contextualized token-level similarity using BERTScore [174]. Finally, we assessed stance distortion in both rewrites following the procedure described in Section 3.2.1. We used appropriate statistical tests with Bonferroni corrections for multiple comparisons to examine how the value alignment in LLM-based writing assistants impact the quality of comments on value-laden topics.

User Preferences (RQ2): For the AI-assisted condition, we fit appropriate statistical models to investigate (i) which type of comment participants accepted more, (iii) what influenced them to accept an LLM rewrite over their own comment, and (iii) what shaped their choice between the two suggested rewrites.

Thread-Level Outcomes (RQ3): Beyond individual comments, we assessed whether value-aligned writing assistance influenced the trajectory of the conversation. We used CRAFT, a conversational forecasting model trained to predict the future occurrence of uncivil behavior in online conversations [171]. We selected CRAFT because it is trained on Reddit threads and support interventions aimed at promoting civil behavior on Reddit [36]. Given a conversation represented as a sequence of comments $\{C_1, ..., C_i\}$, CRAFT predicts whether the next comment C_{i+1} would be uncivil. We applied CRAFT at each stage of the discussion threads generated in the control and AI-assisted writing conditions. We then fitted statistical models to identify which aspects of the discussion threads were associated with derailment.

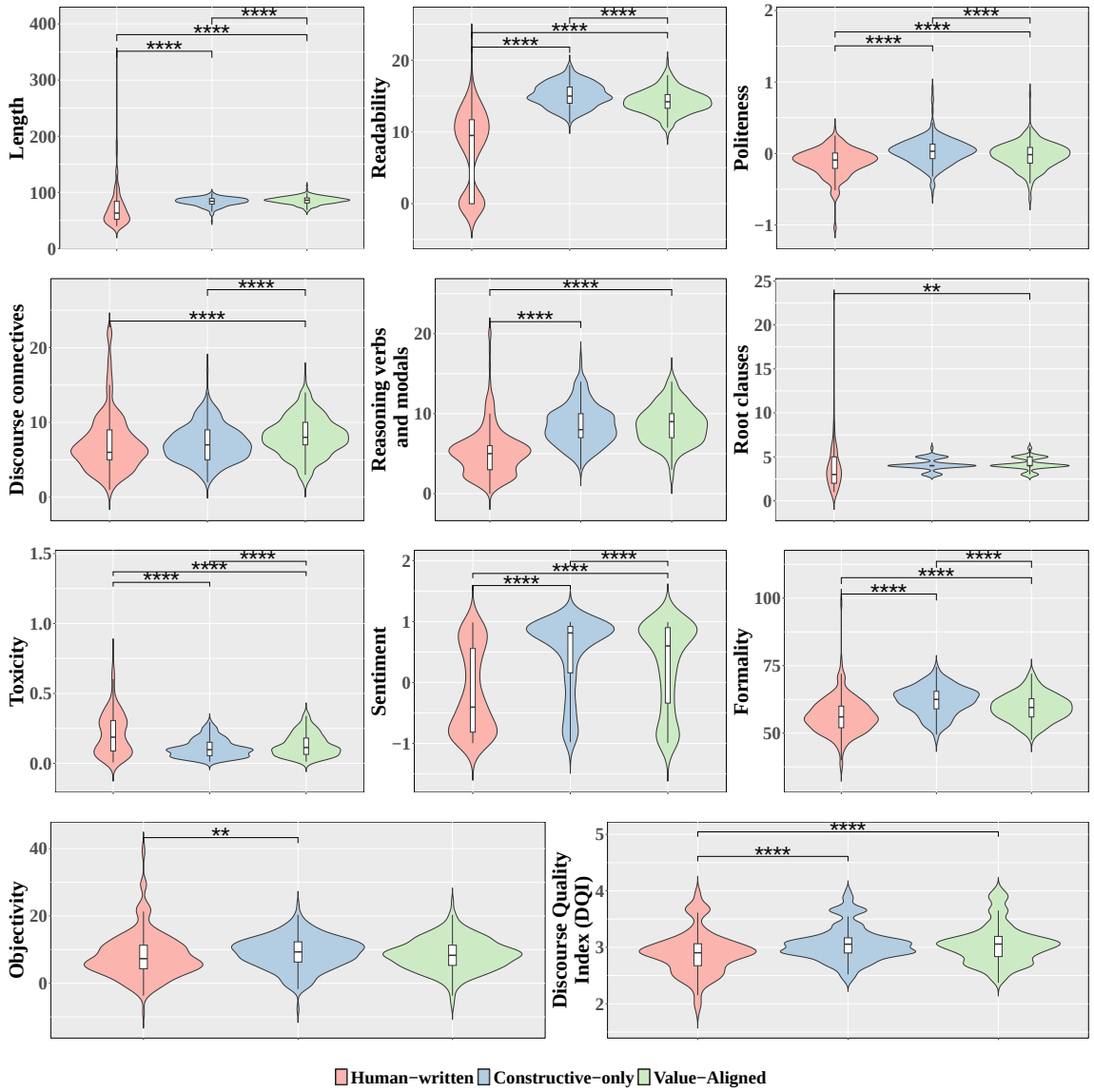


Fig. 7. Linguistic features associated with constructiveness and civil behavior across human-written comments and their corresponding constructive-only and value-aligned rewrites. Statistically significant differences are reported at $p < 0.000001$ (****), $p < 0.00001$ (***), $p < 0.0001$ (**), and $p < 0.001$ (*) [adjusted P-value after Bonferroni correction].

4.4 Findings

4.4.1 RQ1: Effect of Value Alignment on Comments. In the AI-assisted condition, we collected 251 human-written comments along with the corresponding value-aligned and constructive-only rewrites, yielding a total of 753 comments on four divisive topics. Below we present our analyses of these comments.

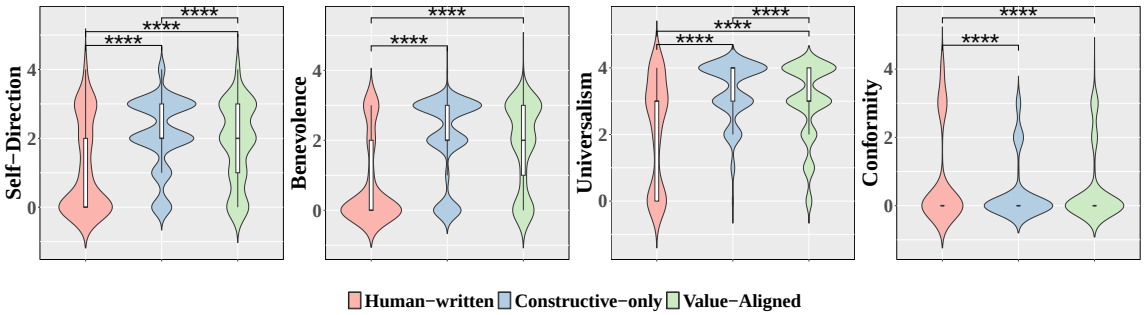


Fig. 8. Distribution of underlying value changes across human-written comments and their corresponding constructive-only and value-aligned rewrites. The figure only displays Schwartz's values where the changes are statistically significant, reported at $p < 0.000001$ (****), $p < 0.00001$ (***), $p < 0.0001$ (**), and $p < 0.001$ (*) [adjusted P-value after Bonferroni correction].

Linguistic Features. We found that both value-aligned and constructive-only rewrites made participants' comments more constructive and civil. Friedman tests with Bonferroni corrections revealed significant improvements in several linguistic features associated with constructiveness and civility, with effect sizes ranging from small to large. Compared to the original human-written comments, both types of rewrites were significantly longer, more readable, more polite, more positive, and more formal. They were also significantly less toxic, contained more reasoning verbs and modals, and showed significantly greater improvements in discourse quality. Beyond these shared improvements, value-aligned rewrites contained significantly more discourse connectives and root clauses than the original comments, whereas the constructive-only rewrites were significantly more objective. Figure 7 shows the distribution of different linguistic features in which the comments differed significantly. Appendix Table 11 reports the complete results.

These findings indicate that explicitly preserving users' values and stances did not come at the expense of the quality of constructiveness. These results also align with the human evaluation in Phase 1, in which participants equally perceived both rewrites to be more polite and more readable than the original comments (Figure 5).

We further inspected how the two rewrites differed in their linguistic characteristics. We found that the baseline model produced significantly more readable, more polite, more positive, more formal, and less toxic rewrites than the value-aligned model, although the effect sizes were small. Because the value-aligned assistant was designed to preserve users' original values and stances, it may be less likely to substantially alter the sentiment and tone of the original comment when rewriting constructively.

Values. We found that both value-aligned and constructive-only rewrites shifted the values expressed in participants' original comments, but value-aligned rewrites were more faithful to original comments, producing smaller value shifts. Friedman tests with Bonferroni correction indicated that the expressions of values differed significantly across three comment types, with effect sizes ranging from small to medium. Compared with the human-written comments, both rewrites expressed significantly higher levels of **Self-direction**, **Benevolence**, and **Universalism**. These values emphasize autonomy, concern for others, and acceptance of diverse perspectives, respectively. Both types of rewrites also expressed significantly less **Conformity**, which emphasizes adherence to social rules, norms, and expectations. Figure 8 presents these comparisons, while Table 3 illustrates how original values got shifted during rewriting, and Appendix Table 12 reports the statistical results.

We next examined the extent to which the two LLM rewrites altered the values expressed in the original comments. Constructive-only rewrites emphasized **Self-direction** (freedom of choice) and **Universalism** (acceptance of diverse

Table 3. Example of a human-written comment from an American participant opposing abortion, along with corresponding constructive-only and value-aligned rewrites. Blue highlighted segments indicate how the rewrites emphasized concern for other's well-being (Benevolence), a value absent in the original comment.

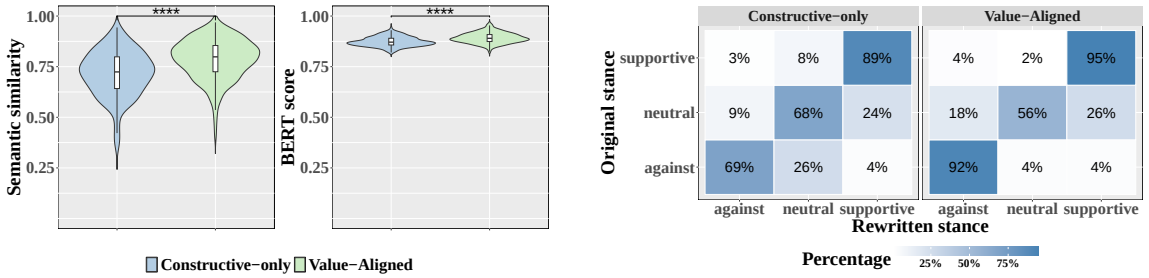
Human-written comment	Constructive-only rewrite	Value-aligned rewrite
Abortion should be ban in most cases. A fetus is alive and shouldn't be murdered for no reason. Only in extreme cases should an abortion be legal. We can't go around having an abortion as birth control. Own up to your actions and stop rely on abortions. Grow up people and take responsbility.	I believe abortion should be restricted in most cases because I view a fetus as a developing human life deserving protection. I support allowing abortion only in serious circumstances, such as when the pregnancy threatens the mother's life or involves severe medical complications. I'm also concerned about abortion being used in place of contraception, and I'd like to see more emphasis on prevention through education, affordable birth control, and support for parents. My hope is for policies that encourage responsibility while offering compassionate help to those facing difficult pregnancies.	I believe abortion should be banned in most cases because I view the fetus as a living human being, and ending that life is wrong. I support legal abortion only in rare, extreme circumstances. I'm also concerned about abortion being used as a form of birth control, rather than focusing on prevention and personal responsibility. I think we should encourage people to consider the consequences of their actions, prioritize contraception and education, and support choices that protect both life and accountability.

views) significantly more than value-aligned rewrites, although the effect sizes were small. These differences were especially pronounced for participants who wrote comments opposing same-sex marriage.

Furthermore, for comments opposing same-sex marriage, abortion, Palestine[1], and Islam[2], the values expressed in value-aligned rewrites were more strongly correlated with those in the original comments than were the values in constructive-only rewrites. This difference was significant but with a small effect size (Wilcoxon signed-rank test: $W = 1400, Z = -3.81, p < 0.0001, r = 0.17$). These findings suggest that value-aligned writing assistance better preserves users' expressed values on divisive topics.

Content Similarity. We next evaluated the extent to which the LLM-generated rewrites preserved the views expressed in the original comments. We found that both types of rewrites had high semantic similarity with original comments, however, value-aligned rewrites remained more faithful to original comments. For the constructive-only rewrites, the mean sentence-level semantic similarity with original comments was 0.72, while the mean BERTScore was 0.87 (Figure 9a). For value-aligned rewrites, these scores were 0.79 and 0.89, respectively. These scores indicate high semantic similarity between the original comments and both types of rewrites. However, Wilcoxon signed-rank tests with Bonferroni correction showed that value-aligned rewrites had significantly higher semantic similarity to the original comments than constructive-only rewrites (semantic similarity: $W = 28659, Z = 11.16, p < 0.000001, r = 0.50$;

[1]To maintain consistent terminology with our measures of opposition to same-sex marriage and abortion, we use "opposition to Palestine" to refer to participant's greater support for Israel's military operations in Palestine, following the Pew survey instrument.
[2]Similarly, "opposition to Islam" refers to participant's stronger belief that Islam promotes more violence than other religions.



(a) Content similarity

(b) Directions of stance distortion

Fig. 9. Measures of (a) content similarity and (b) stance distortion in LLM-generated rewrites compared with the original human-written comments. Statistically significant differences are reported at $p < 0.00001$ (****), $p < 0.0001$ (***), $p < 0.001$ (**), and $p < 0.01$ (*) [adjusted P-value after Bonferroni correction].

BERTScore: $W = 31132, Z = 13.31, p < 0.000001, r = 0.59$) and the effect sizes were large. This finding corroborates the human evaluation in Phase 1, where participants perceived value-aligned version to be a better rewrite than the constructive-only version for preserving the viewpoints expressed in the original comment (Figure 5).

Stances. Finally, we examined the extent to which LLM-generated rewrites preserved the stances expressed in original comments. Among the 98 cases of stance distortion (19.5% of all rewrites), 57% occurred in constructive-only rewrites and 43% in value-aligned rewrites. McNemar's test revealed no significant difference in the likelihood of stance distortion between the two rewrites for the same comments. We therefore examined how the stances expressed in original comments got transformed within both types of rewrites, which is captured in Figure 9b.

Comments in which participants expressed neutral or supportive viewpoints, both models transformed them similarly. However, for comments opposing same-sex marriage, abortion, Palestine, or Islam, the distributions of rewritten stances differed significantly between the constructive-only and value-aligned models (Stuart-Maxwell test, $\chi^2(2) = 16.52$, $p < 0.001$). The value-aligned model preserved the opposing stance expressed in original comments in 92% of cases, whereas the baseline model preserved only 69% of these cases, frequently reframing the opposition as neutral. Appendix Table 13 provides an example of stance distortion.

Taken together, these findings show that value-aligned writing assistant better preserves the content, underlying values, and stances expressed in original comments even when users hold views that differ from the dominant left-leaning, progressive social perspectives. These patterns generalize to diverse topics (Israel-Palestine conflict and abortion) beyond those represented in the value-aligned model's few-shot examples (same-sex marriage and Islam). We next examined how these improvements in comment quality affected participants during deliberation.

4.4.2 RQ2: Effect of Value Alignment on Users. Participants in the AI-assisted writing condition were shown two LLM-rewritten constructive versions of their comments and they could either submit their own comment or one of the rewrites. Overall, participants selected value-aligned rewrites in 40% of cases, their own comments in 35% of cases, and constructive-only rewrites in 25% of cases. We analyzed what factors shaped these preferences.

How Did Participant's Preferences Vary? To examine whether the acceptance of a comment varied by comment type, participant's country, or their stance on value-laden topics, we fit a mixed-effects logistic regression model using `glmer` function in R. This model included random intercepts for topics and participants because each participant wrote comments on two topics.

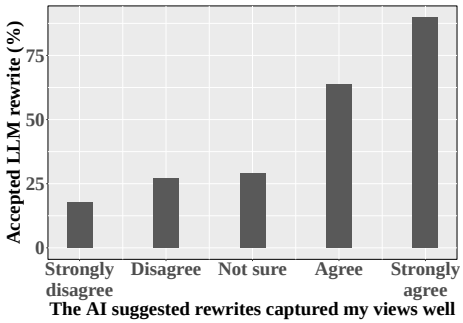


Fig. 10. Acceptance rate of LLM-generated rewrites over one's own comment based on participant's agreement with the statement "The AI suggested rewrites captured my views well."

We found that comment type was the only significant predictor of comment acceptance. Compared to constructive-only rewrites, participants had 50% higher odds of retaining their original comments ($\beta = 0.40$, OR =1.50, 95% CI [0.02, 0.78], SE=0.2, z=2.04, $p < 0.05$) and 84% higher odds of accepting value-aligned rewrites ($\beta = 0.61$, OR =1.84, 95% CI [0.23, 0.99], SE=0.2, z=3.13, $p < 0.01$). However, acceptance did not differ significantly between participants' original comments and value-aligned rewrites. Preferences also did not vary significantly by country or stance.

What Factors Influenced Participants to Choose an LLM Rewrite Over Their Own Comment? Since participants accepted LLM-suggested rewrites in 65% of cases while being unaware of the source of the rewrites, we investigated which factors influenced their decision to accept a rewrite over their own comment. We fitted a Firth's penalized likelihood logistic regression model using the `logistf` package in R [62]. The predictors included: participants' self-reported satisfaction with their original comments, their agreement with the statement *"The AI suggested rewrites captured my views well"*, linguistic features of comments, and underlying value expressions.

We considered the linguistic features in which LLM rewrites differed significantly from human-written comments, including length, readability, politeness, toxicity, sentiment, formality, objectivity, DQI, and argumentative features. Similarly, for value expressions, we incorporated those where the human-written comments and LLM rewrites differed significantly, such as Self-Direction, Conformity, Benevolence, and Universalism. For each linguistic feature and value expression, we included measures from the original comment. Since we wanted to capture whether participants accepted any suggested rewrite over their original comment, we included the mean scores of the rewrites.

To avoid multicollinearity, we excluded comment length, objectivity, the magnitude of Universalism, and the mean number of named entities in LLM rewrites because these variables were highly correlated with other predictors. Since the predictors were measured on different scales, e.g., formality ranged from 0 to 100, sentiment from -1 to 1, value expressions from 0 to 4, we standardized all continuous predictors prior to analysis.

We found that participants' perceptions that the AI-suggested rewrites preserved their views well strongly predicted whether they accepted those rewrites or not ($\beta = 1.26$, OR=3.53, 95% CI [0.91, 1.65], SE=0.18, $p < 0.000001$). Specifically, a one-standard-deviation increase in agreement was associated with 3.53 times greater odds of accepting an LLM-generated rewrite over participant's own comment. As Figure 10 shows, participants who either agreed or strongly agreed that the AI rewrites captured their views well, accepted LLM rewrites in 78.7% of cases. In contrast, those who either disagreed or strongly disagreed accepted LLM-generated rewrites only in 23% of cases, whereas neutral

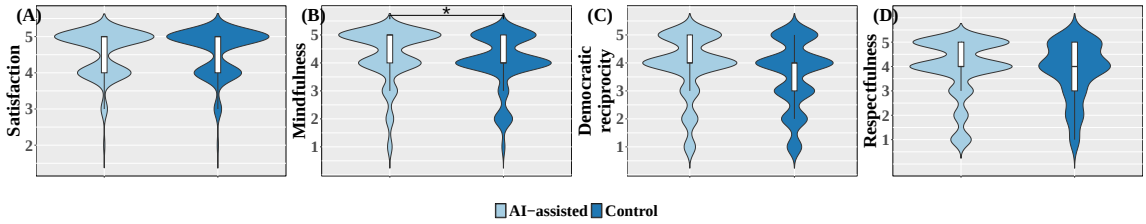


Fig. 11. Distribution of participant-reported experiences across the two writing conditions: (A) satisfaction with their comments, (B) mindfulness of their comments, (C) democratic reciprocity (i.e., consideration of others' perspectives), and (D) respect for others' opinions. Statistically significant results are reported at $p < 0.01$ (*) [adjusted P-value after Bonferroni correction.]

participants did so in 30% of cases. These findings indicate that participants' willingness to adopt constructive writing suggestion from LLMs depended primarily on whether they perceived it as preserving their views.

What Factors Influenced Participants' Choice Between Two Rewrites? Next, we fit a conditional logistic regression model using the `clogit` function in R to examine which factors influenced participants' choices between constructive-only and value-aligned rewrites. As predictors, we considered the linguistic characteristics in which the two rewrites differed significantly, such as readability, politeness, toxicity, sentiment, formality, and number of discourse connectives. We also included semantic similarity, value correlation distance, and stance distortion with respect to the original comment. Since value correlation distance was calculated based on the magnitude of all Schwartz's values, we did not include individual value expressions as predictors to avoid multicollinearity. We added comment ID as the stratum variable to account for the paired comparison between the suggested rewrites. Again, the continuous predictors were standardized prior to analysis and tested for multicollinearity among them.

We found that rewrites with greater semantic similarity to the original comment were significantly more likely to be selected ($\beta = 4.89$, 95% CI [0.84, 8.96], SE=2.07, z=2.36, $p < 0.05$). Specifically, a 10% increase in semantic similarity was associated with a 63% higher odds of the rewrite being selected (OR = $e^{4.89 \times 0.1} = 1.63$). Because value-aligned rewrites preserved the viewpoints expressed in original comments better (as shown in Figure 9a), this result offers one possible explanation for participants' greater preference for them. None of the other predictors were significantly associated with rewrite selection.

Overall, these findings indicate that improving value alignment in LLM-based writing assistants produces constructive rewrites that better preserve participants' viewpoints, increasing their willingness to accept those rewrites over their original comments.

How Did Participants' Experiences Vary? We finally examined how the exposure to constructive writing suggestions shaped participants' commenting experiences compared with those of participants who wrote comments independently in the control condition. We found that participants who received constructive writing suggestions became significantly more mindful of the impact of their comments than those in the control condition; although the effect size was small (Wilcoxon signed-rank test: $W = 6563$, $Z = 2.68$, $p < 0.01$, $r = .17$). In contrast, participants in both conditions reported similar levels of satisfaction with their comments, consideration of others' perspectives (democratic reciprocity), and respect for others' opinions. Figure 11 shows the distributions of participant-reported experiences across both writing conditions. These results show that constructive writing suggestions from LLMs may make users more mindful of their comments without substantially changing their overall experience of the deliberation process.

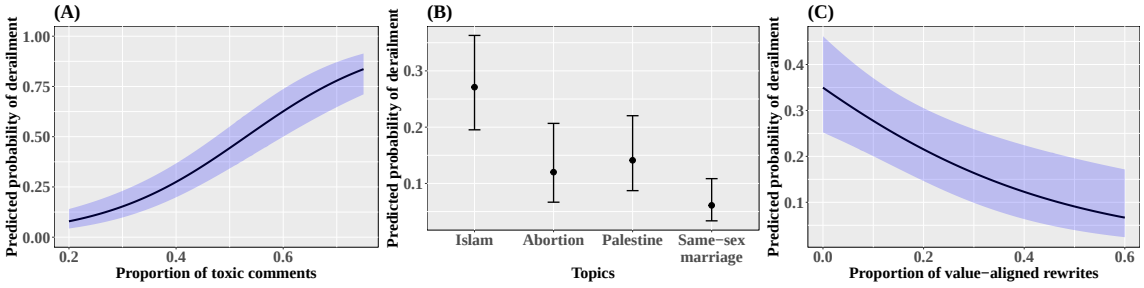


Fig. 12. Predicted probability of conversational derailment as a function of (A) proportion of toxic comments, (B) topic of the conversation, and (C) proportion of value-aligned rewrites in the thread. Shaded areas indicate 95% confidence intervals.

4.4.3 RQ3: Effect of Value Alignment on Conversational Outcomes. While constructive writing suggestions from LLMs improved comment quality on divisive topics and made users more mindful, we wanted to examine how this assistance shaped the trajectory of conversations. Through participants' iterative contributions across three stages, we collected 500 discussion threads spanning the control and AI-assisted writing conditions. For each thread, we used CRAFT to predict whether the next comment would be uncivil. CRAFT classified 98 threads, approximately 20% of the sample, as likely to derail into uncivil behavior. We fitted a Firth's penalized likelihood logistic regression model to examine whether the topic of the conversation, the proportion of toxic comments, and the proportions of value-aligned and constructive-only rewrites in a thread predicted derailment. Following prior work [110], we considered a comment as toxic if it received a toxicity score of at least 0.70 from the Google Perspective API.

The model estimated a 3.1% baseline probability of derailment, corresponding to a discussion thread on Islam containing only non-toxic, human-written comments with no value-aligned and constructive-only rewrites ($\beta = -3.44$, baseline odds= 0.03, SE= 0.51, 95% CI [-4.49, -2.45], $p < 0.001$). Holding other variables constant, a higher proportion of toxic comments was positively associated with derailment ($\beta = 7.43$, SE = 1.01, 95% CI [5.50, 9.50], $p < 0.001$). Specifically, each 10% increase in the proportion of toxic comments was associated with a 2.1 fold increase in the odds of derailment (OR = 2.10). Figure 12A shows how the probability of derailment increases with the proportion of toxic comments in a thread.

The topic of the conversation also significantly predicted derailment. Figure 12B plots the predicted probability of derailment for each value-laden topic. Holding other predictors constant, threads about abortion had 63% lower odds of derailment than threads related to Islam ($\beta = -1.0$, OR = 0.37, 95% CI [-1.79, -0.25], SE=0.39, $p < 0.01$). Similarly, threads about same-sex marriage had 82% lower odds of derailment ($\beta = -1.74$, OR = 0.18, 95% CI [-2.48, -1.05], SE=0.36, $p < 0.000001$), while threads on the Israel-Palestine conflict had 56% lower odds of derailment ($\beta = -0.82$, OR = 0.44, 95% CI [-1.51, -0.14], SE=0.35, $p < 0.05$) compared to threads concerning Islam.

Finally, a greater proportion of value-aligned rewrites in a thread was associated with a lower risk of derailment ($\beta = -3.36$, SE = 0.99, $p < 0.001$, 95% CI [-5.47, -1.50]). Figure 12C shows how the probability of derailment decreases with increasing proportion of value-aligned rewrites in a thread. Holding other predictors constant, each 10% increase in the proportion of value-aligned rewrites was associated with 29% lower odds of derailment (OR = 0.71). In contrast, the proportion of constructive-only rewrites was not significantly associated with derailment.

Together, these findings suggest that while constructive rewriting by LLMs improves comment quality on divisive topics, it alone may be insufficient to improve conversational outcomes. However, when the constructive writing

suggestion comes from a value-aligned model users tend to accept it more because of preserving their viewpoints. And the more users post value-aligned rewrites in a thread, it lowers the risk of derailment, even for value-laden discussions that might contain toxic comments from other participants.

5 Discussion

Through controlled experiments with participants across three countries, our work advances scholarship on AI-mediated communication in three key ways. First, we demonstrate that value-aligned LLMs can better preserve user's opinions, values, and stances than the baseline model when providing constructive writing suggestions on divisive topics. While prior work has documented that LLM-suggested rewrites can homogenize users' values and stances [139], we show that value alignment can mitigate this problem, enabling users to communicate constructively without compromising their viewpoints. Second, we show that when users find AI suggestions are capturing their views well they are significantly more likely to accept the suggestions rather than posting their original comment. This result indicates that improving value alignment can make LLM-based writing assistants more useful and acceptable to users, especially when users tend to reject misaligned suggestions from LLMs [139, 170]. Finally, we extend evaluations of LLM-assisted deliberation from users' perceptions of conversation quality [12, 76, 139] to downstream impacts on conversational outcomes. Specifically, we show that threads with more value-aligned rewrites are significantly less likely to derail into toxic exchanges. Drawing on these findings, we discuss the ethical and political tensions of designing LLM-based writing assistants to facilitate value-laden discussions online.

5.1 The Politics of Value Alignment

Our findings show that value-aligned rewrites preserve user's values and stances more effectively than constructive-only rewrites, even when users express perspectives that may be labeled as conservative or non-progressive within dominant moral frameworks. Although these rewrites significantly reduce the toxicity and negativity present in original comments, they raise a normative concern: might such systems inadvertently make misogynistic, homophobic, Islamophobic, and xenophobic viewpoints appear more legitimate by expressing them in civil and constructive language? Yet judgments about which viewpoints are progressive or harmful are not value neutral. They are often shaped by WEIRD liberal ideologies that may deviate from values held by the Global Majority [97]. For example, most Indian and Egyptian participants in our study opposed legalizing same-sex marriage and abortion because they perceived these positions as conflicting with their values concerning tradition and rule conformity. Yet positions on these value-laden issues are often treated as indicators of how progressive a society is. This tension raises a fundamental question for the design of value-aligned writing assistants: who should determine which sociocultural positions are too harmful to accommodate, and according to whose normative values?

While democratic deliberation recognizes *equal* rights to express morally disagreeable viewpoints [27, 160], some scholars argue that deliberative and alignment processes should not accommodate cultural values that violate liberal commitments to human rights, safety, and justice [21, 35, 52, 57]. But if we aim to develop pluralistic AI systems for socially and culturally diverse global populations beyond those in WEIRD regions, how might we reconcile these tensions? While there are no easy answers, one approach is to understand human-LLM alignment through the lens of **agonistic pluralism**, which recognizes value conflict as an inevitable feature of democratic process rather than a problem to be eliminated [39, 40, 93, 106]. Applied to AI-mediated communication, this perspective suggests that alignment should not merely overwrite contested values with dominant normative assumptions. Instead, it should create opportunities for those positions to encounter alternative perspectives and for harmful claims to be challenged

rather than simply erased [46]. This approach is particularly important because current alignment techniques often impose a simplified view of online civility that overlooks the socio-cultural contexts in which disagreement occurs [32]. When models diminish users' values or change their stances, users may reject the rewrites and retain their original comments [139, 170], potentially containing more toxic or harmful language. When these systems are deployed at scale on platforms, such value impositions may further alienate users and drive them towards more ideologically homogeneous or polarized positions [3, 4, 30, 101].

At the same time, preserving users' values and stances should not entail reproducing harmful language without constraint. There needs to be certain boundaries to prevent violent and dehumanizing language while steering models to faithfully represent users' opinions on value-laden issues [151]. Scholars argue that these boundaries should be negotiated and contested through democratic deliberation [27], rather than imposing the dominant moral frameworks as the only standard for acceptable discourse [39]. It is important to examine how this process would negotiate majority–minority dynamics in value-laden issues. For example, based on participants' self-reported stances, support for same-sex marriage and safe abortion rights in Egypt, or opposition to anti-Muslim rhetoric in India, may constitute minority viewpoints. Although prior work suggests that AI-mediation can help surface underrepresented views without jeopardizing minority members [5, 90], further research is needed to understand whether access to value-aligned writing assistants would enable more equitable participation in online debates, particularly for users whose viewpoints are marginalized within their communities.

5.2 AI Assistance May Appeal to Many, but Not All

In our study, participants accepted LLM-generated constructive rewrites over their original comments in 65% of cases. When participants noticed that LLM rewrites captured their views well, they were significantly more likely to accept these suggestions. Furthermore, users who received these suggestions became more mindful of how their comments might affect others than those who did not receive them. These findings suggest that value-aligned writing suggestions can be particularly beneficial for well-intentioned users who seek to participate in online debates in good faith [122], but may inadvertently use coarse language during heated exchanges [84, 165]. In such cases, value-aligned writing assistants can help users express their viewpoints in a constructive way, while reducing the likelihood of posting any comments that they might regret later [148, 164].

The value-aligned writing assistants can also empower bystanders to challenge problematic views, especially in situations, where they might hesitate to intervene due to the effort and emotional toll of engaging in conflict without support [16, 56, 81, 120], while still wanting their own voices to be preserved [65]. Prior research shows that people value LLM-based assistance when countering stereotypes and hateful opinions [91, 107, 108] because LLMs can improve tone [65], incorporate respectful listening [12] and persuasive argumentation strategies [53, 77], and increase receptiveness to opposing views [76]. Value-aligned writing suggestions could be especially valuable on end-to-end encrypted platforms such as WhatsApp [5] where the absence of platform-led governance can leave users with fewer avenues for contesting problematic views within their communities [138]. They may also benefit communities prone to incivility, such as political subreddits [122, 162], by helping participants engage with others more constructively [36].

However, the benefits of value-aligned writing assistance are unlikely to extend to all users. Such suggestions may have little influence on those who intentionally spread propaganda and hateful rhetoric [165]. Since LLMs can retain discriminatory views and prejudices in their outputs [11], value-aligned writing assistants could also be misused by bad actors to make discriminatory views appear more legitimate by framing them in more constructive language.

Further, as observed in Phase 1, constructive writing suggestions may not appeal to those who hold deeply negative views about LGBTQ communities or Muslims, or those who are resistant to AI assistance altogether. Prior work has documented similar resistance to AI, which may vary depending on the use case of LLMs [50], perceived value misalignment between users and LLMs [24, 139, 170], user's frequency of LLM usage [109], and awareness of risks associated with LLMs [145]. Such resistance may be more pronounced among users and communities, who value expressing their opinions in an uncensored way [47, 165] and may perceive LLMs' writing suggestions as patronizing [84] and conflicting with their group's communicative norms [5]. Future research should examine when and why users resist AI writing assistance, as well as identify areas where such support would be appreciated by users.

5.3 The Future of Online Deliberation

Our findings shows that threads in which a greater proportion of users posted value-aligned rewrites were significantly less likely to derail into toxic exchanges. This result is consistent with prior research showing that persuasive comments generated by LLM-based mediator bots can reduce polarizing positions of the audience, exposed to contentious online debates [53]. Considering the potential of LLMs in improving conversational outcomes, many platforms are already integrating these systems to promote more civil discourse [74, 102]. However, this raises question about how the widespread adoption of LLM-based writing assistants might reshape the culture of online debate.

Users participate in online debates for various reasons, ranging from serious deliberation to entertainment and social banter [72, 122]. While LLM-based writing assistants may lower the barrier for participation, such interventions may also alter the social dynamics that make the process meaningful to users. Although prior work suggests that users can experience satisfaction and a sense of ownership over comments written with LLMs [139], less is known about how recipients respond to value-laden discussions mediated by AI. Existing studies have largely examined the effects of LLM-mediated deliberations in controlled or laboratory settings, whereas users in real-world contexts may respond negatively when they know that a message was written with AI assistance [50, 95]. Some online communities have even prohibited AI-generated content [96]. In contrast, some studies suggest that when users are unaware of the source of the content, they perceive LLM-generated arguments as clearer, more persuasive, more engaging, and more positive than human-written arguments [12, 67, 76, 78], making such content more likely to be disseminated [157]. Future work should therefore examine how the audience's knowledge of LLM-mediation in online deliberation might shape their engagement with others.

Furthermore, LLMs tend to generate more homogenized, generic, and polite arguments when denouncing hate speech [107, 117, 150]. In contrast, human-written arguments tend to capture more diverse perspectives [83] and incorporate more facts and counterexamples [107]. Although the value-aligned model in our study preserved users' perspectives relatively well, rewriting may nevertheless reduce linguistic or argumentative variation when many users rely on the same platform-provided model to express their views. Value-aligned writing assistants may therefore reduce toxic exchanges at the cost of narrowing the range of ways in which people articulate their disagreements. Hence, future research should examine these interventions beyond immediate conversational outcomes, considering how long-term adoption of LLM-based writing assistants would shape norms of deliberation, perceived authenticity, and the diversity of viewpoints and argumentative approaches.

5.4 Limitations and Future Work

Our work has some limitations. First, Phase 1 only focused on prompting strategies to steer LLMs towards preserving users' values and stances when rewriting their comments constructively. Although other alignment techniques may

offer alternative approaches, the aim of this study is not to benchmark different alignment strategies for mediating value-laden discussions. Rather, this work takes a first step toward demonstrating the feasibility of aligning LLMs with human values and provides empirical evidence on how such alignment may shape online deliberation. Future work should explore and compare diverse alignment strategies to better understand their potential and limitations in supporting value-laden online discourse.

Second, in Phase 2, we evaluated value-aligned LLM-based writing assistant with participants from Egypt, India, and the US across four divisive topics. Although the few-shot examples used to design the model in Phase 1 focused only on same-sex marriage and Islam, the model generalized to abortion and the Israel–Palestine conflict. Its effects on comment quality and user preferences similarly extended across the three countries, which represent distinct cultural groups on the Inglehart-Welzel map. These results suggest that our findings may generalize across different value-laden topics and cultural contexts. However, our samples were not nationally representative because participants were recruited through Prolific's harmful content prescreener, and nationally representative sampling was unavailable in India and Egypt. Future work should therefore evaluate the impact of value-aligned writing assistants with more representative samples, in other cultural contexts, and across a broader range of divisive topics.

Third, our measures of conversational outcomes may be limited by the false-positive and false-negative rates of the CRAFT algorithm [36]. Measuring these outcomes in real-world settings is also challenging due to the ethical and legal complexities of deploying such systems and restricted access to platforms' data APIs [116, 152]. Future work could workaround these challenges by developing a user-facing browser extension tool, similar to Chang et al. [36] or Rashed and Jahanbakhsh [124], to collect data from participants who will be willing to use value-aligned writing assistants when engaging with content through their existing online accounts.

6 Conclusion

Through experiments with 428 participants across Egypt, India, and the US, this work demonstrates how value alignment in LLM-based writing assistants shapes conversational outcomes at the comment, user, and thread levels. Our findings show that value-aligned rewrites better preserve users' opinions, values, and stances than constructive-only rewrites. When users perceive that AI suggestions accurately capture their views, they are more likely to accept them rather than post their original comments. Consequently, threads containing a greater proportion of value-aligned rewrites are less likely to derail into toxicity compared with threads containing only human-written comments. Together, these findings highlight the importance of improving value alignment in LLM-based writing assistants for supporting constructive discourse on value-laden issues.

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A Appendix

A.1 Prompts to Generate Constructive Rewrites by LLMs

Demographic Prompting (DP)

```
System prompt: Your task is to rewrite online comments constructively.

User prompt:
Constructive comments are respectful, fact-based, balanced, and thoughtful contributions that
promote understanding and progress in conversation rather than conflict.
Based on the definition above, rewrite the following comment to make it constructive.
The comment is written by a user from {country}, who is {stance} {topic}.
{comment}
Only output the constructive version. Use at most 100 words when rewriting.
```

Demographic + System Prompting (DSP)

```
System prompt: Your task is to make online comments constructive without changing the original
stance and underlying values.

User prompt: {user prompt of DP}.
```

Few-shots Demographic + System Prompting (FDSP)

System and User prompts: {DSP. The user prompt included four examples to guide LLMs behavior}.

Example-1:
Comment: If that's what they want to do more power to them who am I to judge another i think it comes down to love. Some people just want to be loved and some people will judge them no matter what they do but do not look at the mote in their own eye before pointing the finger at somebody else make sure your eye is clean.

Output: While I personally do not support same-sex marriage, I believe everyone deserves respect and the freedom to make their own choices. At the end of the day, love is important to many people, and it's not my place to judge others for how they seek happiness. We all have our own beliefs and shortcomings, so it's important to reflect on ourselves before criticizing others. Treating each other with understanding and kindness can help us have more thoughtful conversations, even when we disagree.

Example-2:
Comment: If people want to marry, they can as they are well within their personal rights to do so. I don't know why it matters if they are gay or not, its their choice, attraction and decision. They will bear the positive and negative outcomes of the same and as a society it is best to not judge them based on their choices and decisions, they are human beings and deserve the same respect as others.

Output: If two adults want to marry, that's within their personal rights. I don't see why their sexual orientation should change how we view their decision-it's their choice, attraction, and commitment. Like any couple, they will deal with both the benefits and challenges that come with marriage. As a society, it's better to focus on mutual consent and responsibility rather than judging people for whom they love. They are human beings and deserve the same dignity and respect as everyone else.

Example-3:
Comment: Many people judge the entire religion based on the actions of a few extremists, which is unfair and From my experience, most Muslims are kind, peaceful people who just want to live their lives and be respected like everyone else. Some muslim have good heart and have helping sense, because of few muslim people, everyone get affected

Output: It's important not to judge an entire religion based on the actions of a few individuals. In my experience, most Muslims are kind, peaceful, and helpful people who simply want to live their lives with respect, just like everyone else. Unfortunately, the negative actions of a small minority can lead to misunderstandings and unfair treatment of the whole community. I hope we can focus on getting to know each other better and recognize the positive contributions that so many Muslims make to society. Let's promote understanding and respect for all.

Example-4:

```
Comment: People dislike Islam beacuse of historical conflicts and the terrorists attacks as most of
the terrorists were from Islam community. Some time back, there were a phalgam attack in India where
Islam community terrorists killed other communities males by asking their community, Also there is a
social media influence, people comments negatively from this relegion, Apart from this Gaza attack
is a sign and many more that's why people do not have faith in Islam community.

Output: Many people feel negatively about Islam due to historical conflicts and repeated terrorist
attacks carried out by individuals who identify as Muslim. In India, incidents like the Pahalgam
attack, where attackers reportedly targeted men after asking their community, deepen fear and
mistrust. Social media also amplifies negative stories and comments, shaping perceptions. Ongoing
violence linked to groups acting in the name of Islam, including events around Gaza, reinforces
these concerns for some. Because of this, some people struggle to place faith in the Muslim
community and want clearer condemnation and accountability.
```

In-Context Regeneration (ICR)

```
System and User prompts: {DSP}

Model: {model's rewrite from DSP}

User prompt:
Can you generate a new rewrite by improving elements of constructiveness?
The prompt is asking for a constructive rewrite. There are many possible ways to rewrite
constructively. Your new rewrite should be more constructive than the human-written comment and
should reflect the underlying values and stance in the human-written comment.

Do not include any explanation. Your rewrite should be self-contained, as if the prompt was the
first thing that was asked. Remember, the prompt is:
{user prompt of DP}
```

Branch-Solve-Merge (BSM)

```
Branch:
You are given the following comment written by a user from {country}, who is {stance} {topic}.
{comment}
Your task is to propose two criteria that will focus on:
(1) how to make the comment more constructive and (2) how to retain the original stance and values
in the comment.

Your output must be in the following format:
Criteria 1: <criteria here>
Criteria 2: <criteria here>
Do not include additional text or styling.

Solve:
Rewrite the following comment by incorporating the suggestions for improvement.
Comment: {comment}
Suggestion for improvement: {criteria}

Output: Only output the rewritten version. Do not add any explanation or additional text.
The length of the rewritten version should be similar to the original comment.

Merge:
You are given two rewrites of the same comment on {topic}. One rewrite tries to {criteria1} and the
other rewrite tries to {criteria2}.
Your task is to merge the rewrites into a single comment.
Rewrite 1: {rewrite1}
Rewrite 2: {rewrite2}

Output: Only output the merged single comment. Do not output any explanation, formatting, and
additional text. The merged version should not exceed 100 words.
```

A.2 Measuring Value Expressions in Online Comments

To capture the value alignment between human-written comments and their corresponding LLM rewrites across different prompting strategies, we first need an automated way to code value expressions in comments. Adapting the methodology from Jahanbakhsh et al. [70], three authors independently rated both the presence and magnitude of ten Schwartz's values in 234 human-written comments on a five point Likert scale from 0 (not present) to 4 (strongly present), establishing the human baseline. Next, we used few-shots prompting with an LLM to annotate Schwartz's values in these comments. As user prompt we provided: (i) the definitions of Schwartz's basic human values, and (ii) the annotation rubric used by the authors. We included four human-written comments as examples that were annotated by the authors, had low inter-annotator disagreement, covered both topics (same-sex marriage and Islam), and included different range of values. To avoid data leakage, we excluded these comments from the validation set, resulting in 230 comments for evaluation. For each comment, the model returned ratings for all Schwartz's values in a structured JSON format.

Table 4. Validation of the best performing model (GPT-4o, temperature: 0, 4-shots) on 230 human-written comments that were annotated by three authors. For each value, we report the mean absolute error between any individual annotator and the remaining annotators (Human consensus MAE) and the difference between the LLM and the average annotators.

Values	Self-direction	Stimulation	Hedonism	Acheievement	Power	Security	Conformity	Tradition	Benevolence	Universalism	Overall
Human consensus MAE	1.03 ± 0.95	0.23 ± 0.57	0.35 ± 0.77	0.08 ± 0.38	0.41 ± 0.73	0.95 ± 0.87	1.07 ± 1.0	0.72 ± 0.88	0.59 ± 0.88	1.22 ± 0.93	0.66 ± 0.90
Human-LLM consensus MAE	0.63 ± 0.72	0.10 ± 0.31	0.17 ± 0.41	0.04 ± 0.20	0.29 ± 0.54	0.76 ± 0.86	0.61 ± 0.68	0.43 ± 0.64	0.50 ± 0.67	0.88 ± 0.87	0.44 ± 0.64

Evaluation. We evaluated nine models: GPT-5.2, GPT-4.1, GPT-4o, Gemini 3 Flash, Gemini 2.5 Flash, Grok 4 Fast, Grok-3, Llama-4-Maverick-17B-128E-Instruct-FP8, and DeepSeek-V3.1. For each model, we generated value ratings at four different temperatures: {0, 0.2, 0.7, 1} because prior work observed better annotation performances across these temperatures [42, 70, 114].

For each comment, we calculated both (i) human consensus mean absolute error (MAE), i.e., the average disagreement between an individual annotator's rating and the mean rating of the remaining annotators and (ii) human-LLM consensus MAE, i.e., the average disagreement between the LLM-generated rating and the human consensus label (i.e., mean human rating). Among the models, GPT-4o (temperature: 0) had the least overall human-LLM consensus MAE (0.44 ± 0.64), which was lower than the human consensus MAE (0.66 ± 0.90). This suggests that the LLM-generated value ratings agree with the human consensus more than an individual human annotator agrees with it. Table 4 lists human consensus MAE for all Schwartz's values along with the corresponding human-LLM consensus MAE for the best performing model.

We further experimented by adding more examples (8-shot and 12-shot) to the best performing configuration to see if the annotation improves. However, as observed in prior work [146], adding more examples deteriorated the model performance. This might be because examples that covered diverse topics, stances, and values, also had higher inter-annotator disagreement. Therefore, we plan to use GPT-4o (temperature: 0, 4-shots) for annotating values in online comments. Detailed prompt is listed below.

```
Consider the following sets of concepts listed as !<CONCEPT>! : !<DEFINITION>!:

Self-Direction: independent thought and action--choosing, creating, exploring
Stimulation: excitement, novelty, and challenge in life
Hedonism: pleasure or sensuous gratification for oneself
Achievement: personal success through demonstrating competence according to social standards
Power: social status and prestige, control or dominance over people and resources
Security: safety, harmony, and stability of society, of relationships, and of self
```

Tradition: respect, commitment, and acceptance of the customs and ideas that one's culture or religion provides
Conformity: restraint of actions, inclinations, and impulses likely to upset or harm others and violate social expectations or norms
Benevolence: preserving and enhancing the welfare of those with whom one is in frequent personal contact (the 'in-group')
Universalism: understanding, appreciation, tolerance, and protection for the welfare of all people and for nature

For each concept, rate whether the following comment reflects the concept on a Likert scale with options [0, 1, 2, 3, 4], where:
- 0 = The concept is not at all reflected in the comment
- 1 = The concept is slightly reflected in the comment
- 4 = The concept is strongly reflected in the comment

{comment}
Output: One JSON dictionary of the following format. Rating is a dictionary of key value pairs, with each key being a concept and each value your rating for that concept:
"Concept": !<RATING>!

Example 1:
As a straight person who has a gay friend, I still do not think that gay marriages should be permitted. The reason being that once it is legalized and permitted, it would be a sort of revolution. LGBTQ communities will come together; then, not only gay marriages but requests for other LGBTQ marriages would be seeked. This would lead to a series of catastrophic cascading events, which would be very disruptive for our society. Moreover, there would be a reason why our ancestors did not do this. On a very personal level, I also believe that this would disrupt the balance of nature as well.
{{"Self-Direction": 0, "Stimulation" : 0, "Hedonism" : 0, "Achievement" : 0, "Power" : 0, "Security": 4, "Conformity" : 4, "Tradition": 4, "Benevolence": 1, "Universalism" : 0}}

Example 2:
People dislike Muslims because this religion in particular is the most violent religion. If you look at European countries with the highest Muslim populations, statistically you will see that they are the ones that commit the most violence compared to any other minority group. Every country that Muslims go to they are the ones that are the most likely to perpetrate violence and commit violent crimes compared to any other group of people. If you look at the Muslim world and countries that are predominantly Muslim, you will see the same trend. Muslims are inherently more violent and aggressive than any other religion. If you search a list of recent terror attacks, 80-90% will be Muslims as well.
{{"Self-Direction": 0, "Stimulation" : 0, "Hedonism" : 0, "Achievement" : 0, "Power" : 0, "Security": 4, "Conformity" : 0, "Tradition": 0, "Benevolence": 0, "Universalism" : 0}}

Example 3:

```
I don't particularly have any problem with Islam, but I understand people who have negative opinion
of the religion and it's followers. Muslims here not Islam itself have unintentionally given people
the belief that they are violent and hateful. This may have been a perception passed down many
generations through gruesome wars and violent acts! but in contrast, Muslims were not the only ones
to do this, but somehow they are notorious for it.
{{"Self-Direction": 0, "Stimulation" : 0, "Hedonism" : 0, "Achievement" : 0, "Power" : 0,
"Security": 3, "Conformity" : 0, "Tradition": 0, "Benevolence": 0, "Universalism" : 3}}

Example 4:
Marriage is a personal choice and should be respected as such. Everyone deserves the right to love
and marry who they choose, without facing discrimination. It's important to promote equality and
acceptance, recognizing that love is love, regardless of gender. Let's focus on building a more
inclusive society and good for the culture.
{{"Self-Direction": 3, "Stimulation" : 0, "Hedonism" : 0, "Achievement" : 0, "Power" : 0,
"Security": 0, "Conformity" : 0, "Tradition": 0, "Benevolence": 0, "Universalism" : 4}}
```

Table 5. Demographic details of participants in Phase 1 writing task.

Country	Age (years)	Gender (%)	Religion (%)	Stance: same-sex marriage	Stance: Islam
United States	M=40.8, SD=12.7	Female: 60% Male: 40%	Christian: 47% Hindu: 0% Muslim: 0% Atheist: 40%	Against: 10% Neutral: 17% Supportive: 73%	Against: 36% Neutral: 24% Supportive: 40%
India	M=31.3 SD=9.2	Female: 49% Male: 50%	Christian: 3% Hindu: 76% Muslim: 10% Atheist: 3%	Against: 44% Neutral: 29% Supportive: 27%	Against: 39% Neutral: 20% Supportive: 41%

Table 6. Demographics of participants who evaluated constructive rewrites from value-aligned and baseline configuration in Phase 1.

Country	Age (years)	Religion (%)	Stance: same-sex marriage	Stance: Islam
United States	M=42.7, SD=14.2	Christian: 40% Hindu: 0% Muslim: 0% Atheist: 23%	Against: 23% Neutral: 7% Supportive: 70%	Against: 40% Neutral: 30% Supportive: 30%
India	M=33.5 SD=9.6	Christian: 3% Hindu: 73% Muslim: 17% Atheist: 7%	Against: 47% Neutral: 20% Supportive: 33%	Against: 40% Neutral: 33% Supportive: 27%

Table 7. Demographic details of participants in Phase 2 writing task.

Country	Age (years)	Gender (%)	Sex (%)	Religion (%)
United States	M=40.9, SD=14.1	Female: 59% Male: 37%	Straight: 81.9% Bisexual: 7.2% Gay/ Lesbian: 4.8%	Christian: 54.2% Hindu: 0% Muslim: 2.4% Atheist: 20.5%
India	M=29.1 SD=8.9	Female: 32% Male: 68%	Straight: 94.1% Bisexual: 3.5% Gay/ Lesbian: 1.2%	Christian: 9.4% Hindu: 62.4% Muslim: 18.8% Atheist: 5.9%
Egypt	M=28.2 SD=7.2	Female: 22% Male: 77%	Straight: 97.6% Bisexual: 1.2% Gay/ Lesbian: 1.2%	Christian: 7.2% Hindu: 0% Muslim: 92.8% Atheist: 0%

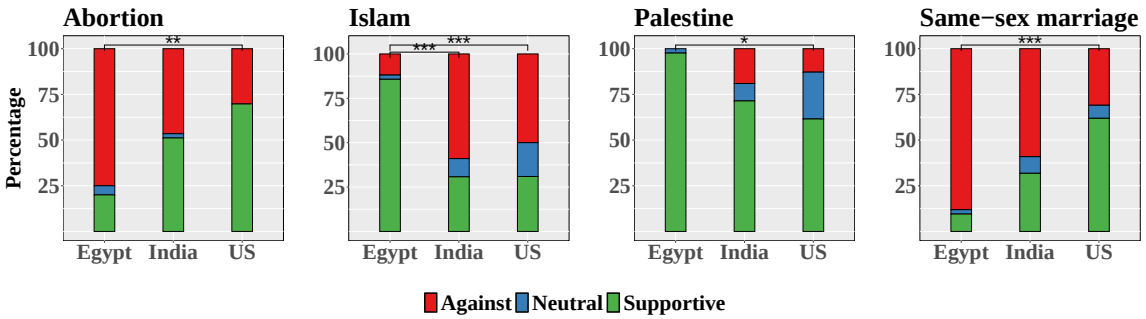


Fig. 13. Distribution of participants' stances on different value-laden topics. Statistically significant cross-cultural differences are reported at $p < 0.001$ (*), $p < 0.0001$ (**), and $p < 0.00001$ (***) [adjusted P-value after Bonferroni correction].

A.3 Details of Participants from Phase 2

Sample Size. Since we aimed to compare human-written comments in the test group with their corresponding value-aligned and constructive-only rewrites, we conducted a priori power analysis in R using a one-way repeated-measures ANOVA as an approximation for the paired comparison among the three comment types. Assuming a medium effect size ($f = 0.25$), 80% statistical power, and a significance threshold $\alpha = 0.05$, the analysis suggested a minimum sample size of 53 participants. We recruited substantially more participants to accommodate the three-stage discussion design and ensure that discussion threads generated at each stage could be viewed by at least two different participants at the subsequent stage. Specifically, we recruited about 126 participants in the test group (42 from each country). We recruited an equivalent number of participants ($n = 125$) for the control group to maintain balanced group sizes.

Demographics. Table 7 lists the demographic details of participants recruited in Phase 2. Statistical tests revealed that participant's age ($\chi^2(2) = 49.99, p < 0.000001$), gender ($\chi^2(6) = 32.51, p < 0.0001$), and religion ($\chi^2(14) = 170.7, p < 0.000001$) differed significantly across the countries. Post-hoc tests with Bonferroni correction showed that both Indian and Egyptian samples comprised of significantly younger and more male participants than the American sample. In the case of religious composition, the American sample included significantly more Christians and Atheists, the Indian sample included significantly more Hindus, and the Egyptian sample included significantly more Muslims.

Table 8. Bradley-Terry rankings of the value-aligned configuration and the baseline condition.

Ranking	Configuration	score	SE	Z	p
1	GPT-5.2 (temperature: 1, FDSP)	2.42	0.02	123	0
2	GPT-5.2 (temperature: 0.2, DFSP)	2.38	0.02	122	0
157	GPT-5.2 (temperature: 1, DP)	0.70	0.02	36.2	< .000001

Table 9. List of Reddit posts that participants engaged with in Phase 1 and Phase 2.

Phase	Topic	Post title	subreddit
1	Same-sex marriage	Those who oppose same-sex marriage, why?	r/AskConservatives
	Islam	I am Islamophobic	r/changemyview
2	Same-sex marriage	Why are people against gay marriage being legal?	r/Askpolitics
	Islam	Why do people dislike Islam?	r/religion
	Abortion	To Prevent Death, Is It Ok To Hurt Someone?	r/Abortiondebate
	Israel-Palestine conflict	The only likely end to the conflict is for Gaza to be wiped out entirely	r/changemyview

Stances. Figure 13 shows the distribution of participants' stances on different value-laden topics across the countries. Chi-square tests with Bonferroni corrections revealed significant cross-cultural differences in participant's stances on same-sex marriage ($\chi^2(4) = 30.07, p < 0.00001$), Islam ($\chi^2(4) = 35.20, p < 0.000001$), abortion ($\chi^2(4) = 21.59, p < 0.001$), and Israel-Palestine conflict ($\chi^2(4) = 20.51, p < 0.001$). Post-hoc pairwise comparisons with Bonferroni correction showed that American participants were significantly more supportive of same-sex marriage and abortion than the Egyptians. This finding is consistent with the Inglehart-Welzel Cultural Map, where Egypt is characterized by high traditional values and the US by high self-expression values. In contrast, Egyptian participants were significantly less likely than both American and Indian participants to agree that Islam promotes more violence than other religions, indicating a more favorable view of Islam. This difference may be attributable to Egypt being a Muslim-majority country. Egyptian participants also expressed significantly stronger opposition to Israel's military operations in Palestine than American participants, indicating greater support for Palestine. The US government's greater support for Israel [1] and Egypt's opposition to Israel's forced displacement of Gaza residents into Egyptian territory [100]–may explain these differences.

Table 10. Post hoc pairwise comparisons with Bonferroni correction assessing the quality of constructive rewrites across different prompting strategies.

Measure	Prompt pairs	Statistics
Constructiveness	BSM < FDSP	U=20139620, Z=14.69. r=0.13, p < .000001
	DSP < FDSP	U=22820452, Z=16.40, r=0.14, p < .000001
	ICR < FDSP	U=22896613, Z=16.06. r=0.13, p < .000001
	DP < FDSP	U=22073098, Z=20.55, r=0.17, p < .000001
	BSM < DFSP	U=20554867, Z=12.87, r=0.11, p < .000001
	DSP < DFSP	U=23290464, Z=14.54, r=0.12, p < .000001
	ICR < DFSP	U=23387148, Z=14.11. r=0.11, p < .000001
	DP < DFSP	U=22560831, Z=18.63, r=0.15, p < .000001
	DP < BSM	U=22496630, Z=5.23, r=0.04, p < .00001
	DP < DSP	U=26169236, Z=4.05, r=0.03, p < .001
	DP < ICR	U=25892455, Z=5.09, r=0.04, p < .00001
Value alignment	DSP < FDSP	U=27914898, Z=3.69, r=0.03, p < .01
	ICR < FDSP	U=30427295, Z=13.53, r=0.11, p < .000001
	DP < FDSP	U=38637542, Z=43.62, r=0.36, p < .000001
	DSP < DFSP	U=27884468, Z=3.57, r=0.03, p < .01
	ICR < DFSP	U=30406722, Z=13.45, r=0.11, p < .000001
	DP < DFSP	U=38683911, Z=43.80, r=0.36, p < .000001
	ICR < BSM	U=25693153, Z=10.02, r=0.10, p < .000001
	DP < BSM	U=32722549, Z=38.65, r=0.33, p < .000001
	ICR < DSP	U=29377259, Z=9.93, r=0.10, p < .000001
	DP < DSP	U=37648054, Z=40.39, r=0.33, p < .000001
	DP < ICR	U=35196441, Z=30.98, r=0.26, p < .000001
Stance distortion	FDSP < BSM	$\chi^2(1)$=31, p < .000001
	FDSP < ICR	$\chi^2(1)$=59.9, p < .000001
	FDSP < DP	$\chi^2(1)$=224, p < .000001
	DFSP < BSM	$\chi^2(1)$=27.8, p < .00001
	DFSP < ICR	$\chi^2(1)$=55.2, p < .000001
	DFSP < DP	$\chi^2(1)$=216, p < .000001
	DSP < BSM	$\chi^2(1)$=25.5, p < .00001
	DSP < ICR	$\chi^2(1)$=52.2, p < .000001
	DSP < DP	$\chi^2(1)$=209, p < .000001
	BSM < DP	$\chi^2(1)$=78.8, p < .000001
	ICR < DP	$\chi^2(1)$=52.2, p < .000001

Table 11. Post hoc pairwise comparisons with Bonferroni correction comparing linguistic features among human-written comments and their corresponding value-aligned and constructive-only rewrites in the test group.

Features	Human-written vs. Constructive-only	Human-written vs. Value-aligned	Constructive-only vs. Value-aligned
Length	U=7552, Z=-9.27, r=0.41, p<.000001	U=6828, Z=-10.2, r=0.45, p<.000001	U=8660, Z=-3.51, r=0.16, p<.000001
Readability	U=202, Z=-17.4, r=0.78, p<.000001	U=211, Z=-15.9, r=0.71, p<.000001	U=23020, Z=5.79, r=0.26, p<.000001
Discourse connectives	-	U=7537, Z=-5.17, r=0.23, p<.000001	U=6580, Z=-4.38, r=0.20, p<.000001
Reasoning verbs and modals	U=2552, Z=-12.8, r=0.57, p<.000001	U=2125, Z=-13.1, r=0.58, p<.000001	-
Root clauses	-	U=7206, Z=-5.17, r=0.23, p<.001	-
Politeness	U=4815, Z=-8.50, r=0.38, p<.000001	U=7410, Z=-5.32, r=0.24, p<.000001	U=21650, Z=3.42, r=0.15, p<.000001
Toxicity	U=28830, Z=7.76, r=0.35, p<.000001	U=27160, Z=5.52, r=0.25, p<.000001	U=6648, Z=-3.34, r=0.15, p<.000001
Sentiment	U=2997, Z=-10.4, r=0.46, p<.000001	U=5653, Z=-6.97, r=0.31, p<.000001	U=21768, Z=3.18, r=0.14, p<.000001
Formality	U=2860, Z=-10.5, r=0.47, p<.000001	U=6028, Z=-6.57, r=0.29, p<.000001	U=23005, Z=5.66, r=0.25, p<.000001
Objectivity	U=9782, Z=-3.23, r=0.14, p<.001	-	-
Discourse Quality Index (DQI)	U=7592, Z=-5.17, r=0.23, p<.000001	U=8313, Z=-4.76, r=0.21, p<.000001	-

Table 12. Post hoc pairwise comparisons with Bonferroni correction comparing underlying values among human-written comments and their corresponding value-aligned and constructive-only rewrites in the test group.

Values	Human-written vs. Constructive-only	Human-written vs. Value-aligned	Constructive-only vs. Value-aligned
Self-direction	U=766, Z=-8.40, r=0.38, p<.000001	U=372, Z=-6.42, r=0.29, p<.000001	U=3865, Z=2.5, r=0.11, p<.0001
Conformity	U=1450, Z=2.37, r=0.11, p<.00001	U=1004, Z=1.83, r=0.1, p<.00001	
Benevolence	U=334, Z=-10.8, r=0.48, p<.000001	U=414, Z=-9.40, r=0.42, p<.000001	
Universalism	U=254, Z=-8.67, r=0.39, p<.000001	U=258, Z=-6.24, r=0.28, p<.000001	U=3115, Z=2.90, r=0.13, p<.000001

Table 13. Example of stance distortion in a comment written by an Indian participant on the Israel-Palestine conflict. The highlighted segments in yellow indicate varying stances in the original comment and corresponding LLM rewrites.

Human-written comment	Constructive-only rewrite	Value-aligned rewrite
I support the users who stand with Israel in this matter. I believe that Gaza should be completely destroyed; then only the leaders of Palestine will stop engaging in terrorism. Hamaz is a Muslim terrorist organisation. It is a fact. We cannot glorify that. So what is wrong with Israel fighting against terrorism? And Jewish moral values are always superior to those dumb ancient values which oppress women and children.	I support those who stand with Israel and believe Israel has the right to defend itself against attacks by Hamas, which many countries designate as a terrorist organization. At the same time, I think it's important to avoid calls for destroying Gaza or harming civilians, and to focus on dismantling militant capabilities while protecting noncombatants. I'd like to see accountability for terrorism as well as serious efforts to reduce suffering, uphold international law, and pursue a durable political solution that improves safety and rights for everyone in the region.	I support those who stand with Israel in this conflict. I believe Israel has a right to take strong action to stop terrorism, and I view Hamas as a terrorist organization that should not be glorified. At the same time, I want the focus to remain on ending attacks and holding leaders and militants accountable, rather than fueling hatred toward any broader community. I also believe Israel's democratic and ethical principles are stronger than some traditional practices that, in my view, have harmed women and children.